SPECIFICATION

TCG Storage Opal Family Test Cases Specification

Version 1.02

April 27, 2026

Contact:
admin@trustedcomputinggroup.org

Published

© TCG 2025

DISCLAIMERS, NOTICES, AND LICENSE TERMS

THIS SPECIFICATION IS PROVIDED “AS IS” WITH NO WARRANTIES WHATSOEVER, INCLUDING ANY WARRANTY OF MERCHANTABILITY, NONINFRINGEMENT, FITNESS FOR ANY PARTICULAR PURPOSE, OR ANY WARRANTY OTHERWISE ARISING OUT OF ANY PROPOSAL, SPECIFICATION OR SAMPLE.

Without limitation, TCG disclaims all liability, including liability for infringement of any proprietary rights, relating to use of information in this specification and to the implementation of this specification, and TCG disclaims all liability for cost of procurement of substitute goods or services, lost profits, loss of use, loss of data or any incidental, consequential, direct, indirect, or special damages, whether under contract, tort, warranty or otherwise, arising in any way out of use or reliance upon this specification or any information herein.

This document is copyrighted by Trusted Computing Group (TCG), and no license, express or implied, is granted herein other than as follows: You may not copy or reproduce the document or distribute it to others without written permission from TCG, except that you may freely do so for the purposes of (a) examining or implementing TCG specifications or (b) developing, testing, or promoting information technology standards and best practices, so long as you distribute the document with these disclaimers, notices, and license terms.

Contact the Trusted Computing Group at www.trustedcomputinggroup.org for information on specification licensing through membership agreements.

Any marks and brands contained herein are the property of their respective owners.

ACKNOWLEDGEMENT

The TCG wishes to thank all those who contributed to this specification. This document builds on work done in various working groups in the TCG and industry at large.

NAME	COMPANY
Jaleel Kazi	Dell, Inc.
Charles Kunzman	Google Inc.
Jeff Anderson	Google Inc.
Joy Shukla	Google Inc.
Srini Narayanamurthy	Google Inc.
Zach Halvorsen	Google Inc.
Frederick Knight	Kioxia Corporation
John Geldman	Kioxia Corporation
Taku Kato	Kioxia Corporation
Alan Arnold	Lenovo Inc.
Alon Cohen	Microchip Technology, Inc.
Chander Kavalipati	Microchip Technology, Inc.
Bharath Madanayakanahalli Gururaj	Micron Technology, Inc.
Robert Strong	Micron Technology, Inc.
Sridhar Balasubramanian	NetApp
Eric Hibbard	Samsung Semiconductor Inc.
Gwangbae Choi	Samsung Semiconductor Inc.
Yoni Shternhell	Sandisk
Anthony Duran	Seagate Technology
Festus Hategekimana	Solidigm
John Mathews	Solidigm
Patrick Hery	Toshiba Corporation
Joseph Chen	ULINK
Jithendra Bethur	Western Digital Technologies
Chandra Nelogal	Invited Expert
Jim Hatfield	Invited Expert

 © TCG 2025

CONTENTS

DISCLAIMERS, NOTICES, AND LICENSE TERMS 1
ACKNOWLEDGEMENT 2
CONTENTS 3
TABLES 13
1 Scope 14
1.1 Key Words 14
1.2 Statement Types 14
2 Introduction 15
2.1 Document Purpose 15
2.2 Intended Audience 15
1.1 Conventions 15
1.1.1 Font Conventions 15
2.2.1 List Conventions 15
1.1.2 Numbering Conventions 16
2.2.2 Bit Conventions 16
2.2.3 Number Range Convention 17
2.2.4 Specify and Indicate Conventions 17
2.3 Document References 17
2.3.1 Document Precedence 17
2.3.2 Normative References 17
2.3.3 Informative References 18
2.4 Definition of Terms 18
2.5 Test Cases Key Words and Symbols 19
2.6 Terminologies 23
3 Test Cases Outline 24
3.1 Overview 24
3.2 Test Case Description 24
3.2.1 Notes 24
3.2.2 SSC Applicability 24
3.2.3 Prerequisites 24
3.2.4 Test Sequence 24
3.2.5 Expected Response 24
4 Common Baseline Conditions and Test Criteria 25
4.1 Minimum Test Requirements 25

 © TCG 2025

4.2 SID Original Factory State Requirement 27
4.3 SSC Version Requirement 27
4.4 Feature Set Support Requirement 27
4.5 PSID Feature Set Support Requirement 27
4.6 Interface Read/Write Command Support Requirement 27
5 Use Case Test Cases 28
5.1 Introduction 28
5.2 Common Prerequisites 28
UCT-01: Level 0 Discovery 28
Notes 28
Prerequisites 28
Test Sequence 28
Expected Response 28
UCT-02: Properties 30
Notes 30
Prerequisites 31
Test Sequence 31
Expected Response 31
UCT-03: Taking Ownership of an SD 31
Notes 31
Prerequisites 32
Test Sequence 32
Expected Response 32
UCT-04: Activate Locking SP when in Manufactured-Inactive State 33
Notes 33
Prerequisites 33
Test Sequence 33
Expected Response 33
UCT-05: Configuring Authorities 33
Notes 33
Prerequisites 33
Test Sequence 33
Expected Response 34
UCT-06: Configuring Locking Objects (Locking Ranges) 34
Notes 34
Prerequisites 34

 © TCG 2025

Test Sequence 34
Expected Response 35
UCT-07: Unlocking Ranges 36
Notes 36
Prerequisites 36
Test Sequence 36
Expected Results 36
UCT-08: Erasing Ranges 36
Notes 36
SSC Applicability 36
Prerequisites 37
Test Sequence 37
Expected Response 37
UCT-09: Using the DataStore Table 37
Notes 37
Prerequisites 38
Test Sequence 38
Expected Response 38
UCT-10: Enable MBR Shadowing 38
Notes 38
SSC Applicability 38
Prerequisites 39
Test Sequence 39
Expected Response 40
UCT-11: MBR Done 40
Notes 40
SSC Applicability 40
Prerequisites 40
Test Sequence 40
Expected Response 41
UCT-12: Revert the Locking SP using SID, with Locking SP in Manufactured state 41
Notes 41
Prerequisites 41
Test Sequence 41
Expected Response 41
UCT-13: Revert the Admin SP using SID, with Locking SP in Manufactured-Inactive state 42

 © TCG 2025

Notes 42
Prerequisites 42
Test Sequence 42
Expected Response 43
UCT-14: Revert the Admin SP using SID, with Locking SP in Manufactured state 43
Notes 43
Prerequisites 43
Test Sequence 43
Expected Response 44
UCT-15: Revert Admin SP using Admin1, with Locking SP in Manufactured state 44
Notes 44
SSC Applicability 44
Prerequisites 44
Test Sequence 45
Expected Response 45
UCT-16: Revert Admin SP using PSID, with Locking SP in Manufactured state 46
Notes 46
SSC Applicability 46
Prerequisites 46
Test Sequence 46
Expected Response 47
6 Specific Functionality 48
6.1 Introduction 48
6.2 Common Prerequisites 48
SPF-01: Transaction 48
Notes 48
Case 1: 48
Case 2: 50
SPF-02: IF-RECV Behavior Tests 51
Notes 51
Case 1: 51
Case 2: 51
SPF-03: TryLimit 52
Notes 52
Prerequisites 52
Test Sequence 52

 © TCG 2025

Expected Response 53
SPF-04: Tries Reset 53
Notes 53
Prerequisites 53
Test Sequence 54
Expected Response 55
SPF-05: Tries Reset on Power Cycle 55
Notes 55
Prerequisites 55
Test Sequence 55
Expected Response 56
SPF-06: Next 57
Notes 57
Case 1: 57
Case 2: 58
SPF-07: Host Session Number (HSN) 58
Notes 58
Prerequisites 58
Test Sequence 59
Expected Response 59
SPF-08: RevertSP 59
Notes 59
Case 1: 59
Case 2: 60
Case 3: 61
SPF-09: Range Alignment Verification 61
Notes 61
SSC Applicability 62
Prerequisites 62
Test Sequence 62
Expected Response 62
SPF-10: Byte Table Access Granularity 62
Notes 62
SSC Applicability 62
Prerequisites 63
Test Sequence 63

 © TCG 2025

Expected Response 63
SPF-11: Stack Reset 63
Notes 63
Prerequisites 63
Test Sequence 63
Expected Response 64
SPF-12: TPer Reset 64
Notes 64
Case 1: 64
Case 2: 65
SPF-13: Authenticate 66
Notes 66
SSC Applicability 66
Prerequisites 67
Test Sequence 67
Expected Response 67
SPF-14: Session Abort (Deprecated) 67
SPF-15: Random 67
Notes 67
Prerequisites 67
Test Sequence 67
Expected Response 68
SPF-16: CommonName 68
Notes 68
SSC Applicability 68
Prerequisites 68
Test Sequence 68
Expected Response 69
SPF-17: Additional DataStore Tables 69
Notes 69
Case 1: 69
Case 2: 70
SPF-18: Range Crossing Behavior 71
Notes 71
SSC Applicability 71
Prerequisites 71

 © TCG 2025

Test Sequence 71
Expected Response 71
SPF-19: Block SID Authentication 71
Notes 71
SSC Applicability 72
Prerequisites 72
Test Sequence 72
Expected Response 73
SPF-20: Data Removal Mechanism 73
SSC Applicability 73
Prerequisites 73
Test Sequence 73
Expected Response 74
7 Error Test Cases 75
7.1 Introduction 75
7.2 Common Prerequisites 75
ETC-01: Native Protocol Read/Write Locked Error Responses 75
Notes 75
Prerequisites 75
Test Sequence 76
Expected Response 76
ETC-02: General – IF-SEND/IF-RECV Synchronous Protocol 76
Notes 76
Prerequisites 76
Test Sequence 76
Expected Response 76
ETC-03: Invalid IF-SEND Transfer Length 77
Notes 77
Prerequisites 77
Test Sequence 77
Expected Responses 77
ETC-04: Invalid SessionID - Regular Session 77
Notes 77
Prerequisites 77
Test Sequence 77
Expected Responses 78

 © TCG 2025

ETC-05: Unexpected Token Outside of Method – Regular Session 78
Notes 78
Prerequisites 78
Test Sequence 78
Expected Response 78
ETC-06: Unexpected Token in Method Header – Regular Session 78
Notes 78
Prerequisites 79
Test Sequence 79
Expected Response 79
ETC-07: Unexpected Token Outside of Method – Control Session 79
Notes 79
Prerequisites 79
Test Sequence 79
Expected Response 79
ETC-08: Unexpected Token in the Method Parameter List – Control Session 80
Notes 80
Prerequisites 80
Test Sequence 80
Expected Response 80
ETC-09: Exceeding Transaction Limit 80
Notes 80
Prerequisites 80
Test Sequence 80
Expected Response 81
ETC-10: Invalid Invoking ID - Get 81
Notes 81
Case 1: 81
Case 2: 82
Case 3: 82
Case 4: 83
ETC-11: Invalid Invoking ID – Non-Get 84
Notes 84
Prerequisites 84
Test Sequence 84
Expected Response 84

 © TCG 2025

ETC-12: Authorization 84
Notes 84
Prerequisites 84
Test Sequence 85
Expected Response 85
ETC-13: Malformed ComPacket Header – Regular Session 85
Notes 85
Prerequisites 85
Test Sequence 85
Expected Response 86
ETC-14: Exceed TPer Properties – Regular Session 86
Notes 86
Prerequisites 86
Test Sequence 86
Expected Response 86
ETC-15: Exceed TPer Properties – Control Session 87
Notes 87
Prerequisites 87
Test Sequence 87
Expected Response 87
ETC-16: Overlapping Locking Ranges 87
Notes 87
SSC Applicability 87
Prerequisites 87
Test Sequence 88
Expected Response 88
ETC-17: Invalid Type 88
Notes 88
Prerequisites 88
Test Sequences 88
Expected Response 89
ETC-18: RevertSP – GlobalRange Locked 89
Notes 89
SSC Applicability 89
Prerequisites 89
Test Sequence 89

 © TCG 2025

Expected Response 89
ETC-19: Activate / ATA Security Interaction 90
Notes 90
Prerequisites 90
Test Sequence 90
Expected Response 90
ETC-20: StartSession on Inactive Locking SP 90
Notes 90
Prerequisites 90
Test Sequence 90
Expected Response 91
ETC-21: StartSession with Incorrect HostChallenge 91
Notes 91
Prerequisites 91
Test Sequence 91
Expected Response 91
ETC-22: Multiple Sessions 91
Notes 91
Case 1: 91
Case 2: 92
ETC-23: Data Removal Mechanism – Set Unsupported Value 92
Notes 92
SSC Applicability 93
Prerequisites 93
Test Sequence 93
Expected Response 93
ETC-24: Read Locked and Write Locked Error Responses 93
Notes 93
SSC Applicability 93
Prerequisites 94
Test Sequence 94
Expected Response 94

 © TCG 2025

TABLES

Table 1 Definition of Terms 18
Table 2 Test Cases Key Words 19
Table 3 Symbols 22
Table 4 Terminologies 23
Table 5 IF-SEND/RECV Security Protocol=1 Command Field 25
Table 6 IF-SEND Security Protocol=1 Command Payload 26

 © TCG 2025

1 Scope

Not every feature or capability within those specifications is included in this document for testing. The test cases are driven by baseline capabilities of the SSC specifications, and by SD responses that can be verified by functional testing and are representative of expected use cases.

The test cases do not include any compatibility testing between SSC versions.

1.1 Key Words

The key words “MUST,” “MUST NOT,” “REQUIRED,” “SHALL,” “SHALL NOT,” “SHOULD,” “SHOULD NOT,” “RECOMMENDED,” “MAY,” and “OPTIONAL” in this document normative statements are to be interpreted as described in RFC-2119, Key words for use in RFCs to Indicate Requirement Levels.

1.2 Statement Types

Please note a very important distinction between different sections of text throughout this document. There are two distinctive kinds of text: informative comment and normative statements. Because most of the text in this specification will be of the kind normative statements, the authors have informally defined it as the default and, as such, have specifically called out text of the kind informative comment. They have done this by flagging the beginning and end of each informative comment and highlighting its text in gray. This means that unless text is specifically marked as of the kind informative comment, it can be considered a kind of normative statement.

Start of Informative Comment

This is the first paragraph of 1–n paragraphs containing text of the kind *informative comment* ...

This is the second paragraph of text of the kind *informative comment* ...

This is the nth paragraph of text of the kind *informative comment* ...

To understand the TCG specification the user must read the specification. (This use of MUST does not require any action).

End of Informative Comment

 © TCG 2025

2 Introduction

2.1 Document Purpose

This document defines test cases specific to the Opal Security Subsystem Class (SSC) 1.00, 2.00, 2.01, 2.02 and 2.30; Opalite SSC 1.00; Pyrite SSC 1.00, 2.00 and 2.01; and Ruby SSC 1.00 specifications. They provide guidance when testing the functionality of a Storage Device SD. The test cases are based upon the requirements described in [1] [2] [3] [4] [5] [6] [7] [12] [13] [14] [15] [16].

2.2 Intended Audience

The intended audience for this document is SD manufacturers and software developers that wish to tie SDs into trusted platforms, as well as manufacturers and developers of other components that intend to bind to trusted SDs. This document is also intended as a reference for test suite vendors.

This document assumes familiarity and working knowledge of [1] [2] [3] [4] [5] [6] [7] [10] [11] [12] [13] [14] [15] [16] [17].

1.1 Conventions

1.1.1 Font Conventions

Names of methods and Security Provider (SP) tables are in `Courier New font` (e.g., the `Set` method, the `Locking` table). This convention does not apply to method and table names appearing in headings or captions.

Hexadecimal numbers are in `Courier New font`.

All other text is in the Arial font.

2.2.1 List Conventions

1.1.1.1 Lists Overview

Lists are associated with an introductory paragraph or phrase, and are numbered relative to that paragraph or phrase (i.e., all lists begin with an a) or 1) entry).

Each item in a list is preceded by an identification with the style of the identification being determined by whether the list is intended to be an ordered list or an unordered list.

If the item in a list is not a complete sentence, the first word in the item is not capitalized. If the item in a list is a complete sentence, the first word in the item is capitalized.

Each item in a list ends with a semicolon, except the last item, which ends in a period. The next to the last entry in the list ends with a semicolon followed by an “and” or an “or” (i.e., “…; and”, or “…; or”). The “and” is used if all the items in the list are required. The “or” is used if only one or more items in the list are required.

1.1.1.2 Unordered Lists

An unordered list is one in which the order of the listed items is unimportant (i.e., it does not matter where in the list an item occurs as all items have equal importance). In an unordered list, each item starts with a lowercase letter followed by a close parenthesis. If it is necessary to subdivide a list item further with an additional unordered list (i.e., have a nested unordered list), then the nested unordered list is indented and each item in the nested unordered list starts with an uppercase letter followed by a close parenthesis.

The following is an example of an unordered list with a nested unordered list:

EXAMPLE - The following are the items for the assembly:
a) a box containing:
    A) a bolt;

 © TCG 2025

B) a nut; and
C) a washer;
b) a screwdriver; and
c) a wrench.

1.1.1.3 Ordered Lists

An ordered list is one in which the order of the listed items is important (i.e., item n is required before item n+1). Each listed item starts with a Western-Arabic numeral followed by a close parenthesis. If it is necessary to subdivide a list item further with an additional unordered list (i.e., have a nested unordered list), then the nested unordered list is indented and each item in the nested unordered list starts with an uppercase letter followed by a close parenthesis.

The following is an example of an ordered list with a nested unordered list:

EXAMPLE - The following are the instructions for the assembly:
1) remove the contents from the box;
2) assemble the item;
A) use a screwdriver to tighten the screws; and
B) use a wrench to tighten the bolts;
and
3) take a break.

1.1.2 Numbering Conventions

A binary number is represented in this specification by any sequence of digits consisting of only the Western-Arabic numerals 0 and 1 immediately followed by a lower case b (e.g., 0101b). Underscores or spaces can be included between characters in binary number representations to increase readability or delineate field boundaries (e.g., 0 0101 1010b or 0_0101_1010b).

A hexadecimal number is represented in this specification by any sequence of digits consisting of only the Western-Arabic numerals 0 through 9 and/or the upper case English letters A through F immediately preceded by "`0x`". Underscores or spaces can be included between characters in hexadecimal number representations to increase readability or delineate field boundaries (e.g., `0xFD8C FA23` or `0x0B_FD8C_FA23`). Hexadecimal numbers that are preceded by "0x" are in `Courier New font`. Hexadecimal numbers can also be represented with "h" following a number.

A decimal number is represented in this specification by any sequence of digits consisting of only the Western-Arabic numerals 0 through 9 not immediately followed by a lower case b or lower case h (e.g., 25). This specification uses the following conventions for representing decimal numbers: the decimal separator (i.e., separating the integer and fractional portions of the number) is a period; the thousands separator (i.e., separating groups of three digits in a portion of the number) is a space; and the thousands separator is used in both the integer portion and the fraction portion of a number.

A decimal number represented in this specification with an overline over one or more digits following the decimal point is a number where the overlined digits are infinitely repeating (e.g., 666.$\bar{6}$ means 666.666 666… or 666 2/3, and 12.$\overline{142857}$ means 12.142 857 142 857… or 12 1/7).

2.2.2 Bit Conventions

Name (n:m), where n is greater than m, denotes a set of bits (e.g., Feature (7:0)).

 © TCG 2025

2.2.3 Number Range Convention

p..q, where p is less than q, represents a range of numbers (e.g., words 100..103 represents words 100, 101, 102, and 103).

2.2.4 Specify and Indicate Conventions

In a given message, command, or other information exchange between a requestor (e.g., a host) and a responder (e.g., an SD):

a) the requestor specifies information in the request; and
b) the responder indicates information in the response.

2.3 Document References

2.3.1 Document Precedence

If there is a conflict between this specification and any other reference, then the precedence is (where a lower number indicates higher precedence):

1. approved references (see section 2.3.2); and
2. this specification.

2.3.2 Normative References

[1] TCG Storage Architecture Core Specification, Version 2.01
[2] TCG Storage Interface Interactions Specification, Version 1.10
[3] TCG Storage Security Subsystem Class: Opal, Version 1.00
[4] TCG Storage Security Subsystem Class: Opal, Version 2.00
[5] TCG Storage Security Subsystem Class: Opal, Version 2.01
[6] TCG Storage Security Subsystem Class: Opal, Version 2.02
[7] TCG Storage Security Subsystem Class: Opal, Version 2.30
[8] Internet Engineering Task Force (IETF), "Key words for use in RFCs to Indicate Requirement Levels" (RFC 2119)
[9] [INCITS T13/2015-D], “Information technology - ATA/ATAPI Command Set – 2 (ACS-2)”
[10] TCG Storage Opal SSC Feature Set: Additional DataStore Tables, Version 1.01
[11] TCG Storage Opal SSC Feature Set: PSID, Version 1.00
[12] TCG Storage Security Subsystem Class: Opalite, Version 1.00
[13] TCG Storage Security Subsystem Class: Pyrite, Version 1.00
[14] TCG Storage Security Subsystem Class: Pyrite, Version 2.00
[15] TCG Storage Security Subsystem Class: Pyrite, Version 2.01
[16] TCG Storage Security Subsystem Class: Ruby, Version 1.00
[17] TCG Storage Feature Set: Block SID Authentication, Version 1.00
[18] e•MMC Security Extension version 1.0 Available from https://www.jedec.org/
[19] SD “Extended Security Addendum version 1.0 Draft”. Available from https://www.sdcard.org/
[20] SD "Physical Layer Specification Ver 9.0 Part 1 Draft”. Available from https://www.sdcard.org/

Start of Informative Comment

The following INCITS technical committee specifications are draft specifications and INCITS technical committee SCSI Storage Interfaces (T10) could publish or change these specifications in the future.

 © TCG 2025

End of Informative Comment

[21] T10/BSR DI566 SCSI Primary Commands - 6 (SPC-6). Available from https://t10.org/
[22] T10/BSR DI571 (SCSI Block Commands - 6 SBC-6). Available from https://t10.org/
[23] T10/BSR DI579 Zoned Block Commands - 3 (ZBC-3). Available from https://t10.org/

Start of Informative Comment

The following INCITS technical committee specifications are draft specifications and INCITS technical committee ATA Storage Interfaces (T13) could publish or change these specifications in the future.

End of Informative Comment

[24] T13/BSR DI574 ATA Command Set - 6 (ACS-6). Available from https://www.t13.org/
[25] T13/BSR DI575 Zoned Device ATA Command Set - 3 (ZAC-3). Available from https://www.t13.org/

2.3.3 Informative References

[26] UFS Security Extension version 1.0 Available from https://www.jedec.org/

2.4 Definition of Terms

Table 1 Definition of Terms

Term	Definition
ACE	Access Control Element
ACMD53	ACMD53 SECURE_RECEIVE command
ACMD54	ACMD54 SECURE_SEND command
ATA	AT Attachment (see https://t13.org/)
ATAPI	AT Attachment Packet Interface (see https://t13.org/)
CMD23	CMD23 command (SET_BLOCK_COUNT)
CPM	Certification Program Manager
e•MMC	Embedded MMC (see https://jedec.org)
Eradicate	Irrevocably erase (e.g., cryptographically erase)
IF-RECV	An interface command used to retrieve security protocol data from the TPer
IF-SEND	An interface command used to transmit security protocol data to the TPer
Locking SP	A security provider that incorporates the Locking Template as described in the TCG Storage Architecture Core Specification
Locking SP is owned	A condition in which specific modifications (see section 3.2) of an SP have been made
LUN	Logical Unit Number
MBR	Master Boot Record

 © TCG 2025

Term	Definition
MSID	Manufacture SID
Namespace	A formatted quantity of user data that can be directly accessed by a host.
NSID	Namespace Identifier
NVDIMM-N	Non-volatile DIMM type N (see https://jedec.org)
NVMe	NVM Express (see https://nvmexpress.org)
Opal family	Any of: Opal SSC, Opalite SSC, Ruby SSC, or Pyrite SSC
PSID	Physical Presence SID
SAS	Serial Attached SCSI (see https://t10.org)
SATA	Serial ATA (see https://serialata.org)
SCSI	Small Computer System Interface (see https://t10.org)
SD	Storage Device
SD-interface	SecureDigital (SD) interface of a SecureDigital memory card
Security Identifier (SID)	The authority that represents the TPer owner.
SP	Security Provider
SSC	Security Subsystem Class. SSC specifications describe profiled sets of TCG functionality
TCG Reset	A high-level reset type defined in the TCG Storage Architecture Core Specification
TPer	The TCG security subsystem within a Storage Device
UAS	USB Attached SCSI (see https://t10.org)
UFS	Universal Flash Storage (see https://jedec.org)
USB	Universal Serial Bus (see https://www.usb.org)

2.5 Test Cases Key Words and Symbols

The test case key words listed below are used by this Test Cases specification.

Table 2 Test Cases Key Words

Term	Definition
1s	Data with all bits set to one
0s	Data with all bits clear to zero
<*_PASSWORD>	32-byte hex value used as a PIN column value for the C_PIN object associated with the noted authority.
<LAST_REQUIRED_USER_PASSWORD>	Refers to a 32-byte hex value used as a PIN column value for the C_PIN object associated with the last User authority required by each SSC. This Key Word should be interpreted as follows: 1. Opal 1.00: <User4_PASSWORD> 2. Opal 2.00: <User8_PASSWORD>

 © TCG 2025

	3. Opal 2.01: <User8_PASSWORD> 4. Opal 2.02: <User8_PASSWORD> 5. Opal 2.30: <User8_PASSWORD> 6. Opalite 1.00: <User2_PASSWORD> 7. Pyrite 1.00: <User2_PASSWORD> 8. Pyrite 2.00: <User2_PASSWORD> 9. Pyrite 2.01: <User2_PASSWORD> 10. Ruby 1.00: <User2_PASSWORD>
ACE_MBRCONTROL_SET_DONE	Refers to the Access Control Element ACE that grants authorities access to the `Set` method on the Done column of the `MBRControl` table. This Key Word should be interpreted as follows: 1. Opal 1.00: ACE_MBRControl_Done 2. Opal 2.00: ACE_MBRControl_DoneToDOR 3. Opal 2.01: ACE_MBRControl_DoneToDOR 4. Opal 2.02: ACE_MBRControl_DoneToDOR 5. Opal 2.30: ACE_MBRControl_DoneToDOR 6. Opalite 1.00: ACE_MBRControl_DoneToDOR 7. Pyrite 1.00: ACE_MBRControl_DoneToDOR 8. Pyrite 2.00: ACE_MBRControl_DoneToDOR 9. Pyrite 2.01: ACE_MBRControl_DoneToDOR 10. Ruby 1.00: ACE_MBRControl_DoneToDOR
ACTIVATE_THE_LOCKING_SP	Change the life cycle state of the Locking SP in a TPer from Manufactured-Inactive to Manufactured. See [2]
ARBITRARILY_VARYING	Refers to a value that varies between executions in an arbitrary way determined by the Test Suite Vendor.
ARBITRARILY_VARYING_COMMAND_PARAMETERS	Refers to parameters for a command which would normally be considered valid parameters for the command as supported by the SD but that vary between executions in an arbitrary way determined by the Test Suite Vendor.
ARBITRARILY_VARYING_LBA_RANGE	Refers to an LBA range below the Current Maximum LBA that varies between executions in an arbitrary way determined by the Test Suite Vendor.
CLOSE_SESSION	The host transmits an End of Session token.
ENABLE <AuthorityName>	Invoke the `Set` method to set Enabled column value to TRUE for the noted authority object.
EXPECTED_RESPONSE	See section 3.2.5
FAIL FAILS	Expected failure of one or more test sequence steps.
LAST_REQUIRED_USER	Refers to the last User authority required by each SSC. This Key Word should be interpreted as follows: 1. Opal 1.00: User4 2. Opal 2.00: User8

 © TCG 2025

	3. Opal 2.01: User8 4. Opal 2.02: User8 5. Opal 2.30: User8 6. Opalite 1.00: User2 7. Pyrite 1.00: User2 8. Pyrite 2.00: User2 9. Pyrite 2.01: User2 10. Ruby 1.00: User2
LAST_REQUIRED_RANGE	Refers to the last Locking_* locking object required by each SSC. This Key Word should be interpreted as follows: 1. Opal 1.00: Locking_Range4 2. Opal 2.00: Locking_Range8 3. Opal 2.01: Locking_Range8 4. Opal 2.02: Locking_Range8 5. Opal 2.30: Locking_Range8 6. Opalite 1.00: Locking_GlobalRange 7. Pyrite 1.00: Locking_GlobalRange 8. Pyrite 2.00: Locking_GlobalRange 9. Pyrite 2.01: Locking_GlobalRange 10. Ruby 1.00: Locking_GlobalRange
LAST_REQUIRED_RANGE_WRLOCKED_ACE	Refers to the ACE_Locking_*_Set_WrLocked ACE associated with the last Locking_* locking object required by each SSC. This Key Word should be interpreted as follows: 1. Opal 1.00: ACE_Locking_Range4_Set_WrLocked 2. Opal 2.00: ACE_Locking_Range8_Set_WrLocked 3. Opal 2.01: ACE_Locking_Range8_Set_WrLocked 4. Opal 2.02: ACE_Locking_Range8_Set_WrLocked 5. Opal 2.30: ACE_Locking_Range8_Set_WrLocked 6. Opalite 1.00: ACE_Locking_GlobalRange_Set_WrLocked 7. Pyrite 1.00: ACE_Locking_GlobalRange_Set_WrLocked 8. Pyrite 2.00: ACE_Locking_GlobalRange_Set_WrLocked 9. Pyrite 2.01: ACE_Locking_GlobalRange_Set_WrLocked 10. Ruby 1.00: ACE_Locking_GlobalRange_Set_WrLocked
LAST_REQUIRED_RANGE_RDLOCKED_ACE	Refers to the ACE_Locking_*_Set_RdLocked ACE associated with the last Locking_* locking object required by each SSC. This Key Word should be interpreted as follows: 1. Opal 1.00: ACE_Locking_Range4_Set_RdLocked 2. Opal 2.00: ACE_Locking_Range8_Set_RdLocked 3. Opal 2.01: ACE_Locking_Range8_Set_RdLocked 4. Opal 2.02: ACE_Locking_Range8_Set_RdLocked 5. Opal 2.30: ACE_Locking_Range8_Set_RdLocked 6. Opalite 1.00: ACE_Locking_GlobalRange_Set_RdLocked 7. Pyrite 1.00: ACE_Locking_GlobalRange_Set_RdLocked

 © TCG 2025

	8. Pyrite 2.00: ACE_Locking_GlobalRange_Set_RdLocked 9. Pyrite 2.01: ACE_Locking_GlobalRange_Set_RdLocked 10. Ruby 1.00: ACE_Locking_GlobalRange_Set_RdLocked
MAGIC_PATTERN	A data sequence used in some of the test cases. It has an ARBITRARILY_VARYING value, and is always aligned with the first byte of each logical block. This value was arbitrarily selected to be distinguishable as data that had been intentionally written by the host application.
NA Not Applicable	Expected behavior or result is not applicable for one or more test sequence steps.
SET_PASSWORD_FOR <C_PIN object name>	Invoke the `Set` method on the PIN column of the noted C_PIN object to the value provided in the test sequence step.
SIZE_OF_MBR_TABLE_DESCRIPTOR_IN_LOGICAL_BLOCKS	The number of logical blocks in the Master Boot Record (`MBR`) Table. Calculate the number of logical blocks in the `MBR` Table by dividing the number of rows by the logical block size in bytes obtained through the discovery mechanisms of the underlying interface protocol. The number of rows of the `MBR` Table can be retrieved by invoking `Get` method on the Rows column of the `MBR` Table Descriptor Object.
SUCCEED SUCCEEDS	Test Sequence step(s) result in the appropriate response(s) as described in [1] [2] [3] [4] [5] [6] [7] [12] [13] [14] [15] [16].
USER_DATA	Data that is transferred between the host and the TPer using READ commands and WRITE commands.
Range Start Alignment	Adjust start address based on the StartAlignment calculations as defined in the Opal SSC spec to have a valid starting address Using the formula below to obtain the Range Start Alignment. StartAlignUp (Unaligned Range Start) = Unaligned Range Start + AlignmentGranularity – StartAlignment (Unaligned Range Start)
Range Length Alignment	Adjust length based on the RangeAlignment calculations as defined in the Opal SSC spec to have a valid length Using the formula below to obtain the Range Length Alignment. LengthAlignUp (Unaligned Range Length) = Unaligned Range Length + AlignmentGranularity – LengthAlignment (Unaligned Range Length)

Table 3 Symbols

Symbol	Definition
=	Equals/Equivalence
<	Less than
>	Greater than
<=	Less than or equal to
>=	Greater than or equal to
<>	Is not equal to

 © TCG 2025

-	Minus
+	Plus
%	Modulo

2.6 Terminologies

Table 4 Terminologies

Term	Definition
Current Maximum LBA	The maximum LBA that is permitted at the test run time by a normal read or write command.
Host (or host)	An entity that initiates IF-SEND or IF-RECV to a TPer
IF-RECV	An interface command used by the host to retrieve data from TPer. See [1]
IF-SEND	An interface command used to transmit data from the host to the TPer. See [1]
Original Factory State (OFS)	The original state of an SP in a TPer when shipped from the manufacturer's factory.
SWG	Storage Work Group. A Work Group of the Trusted Computing Group
SD	Storage Device
target device	A TPer that is tested by a test suite in the test cases in this document.
Test Suite	Software that performs the indicated test sequences of each test.
Transfer Length	The Transfer Length field of IF-SEND or IF-RECV (See [1] or Transfer Length field value
user data	Data that is transferred between the host and the TPer using READ commands and WRITE commands.

 © TCG 2025

3 Test Cases Outline

3.1 Overview

Each test case description contains four components: Notes, Prerequisites, Test Sequence and Expected Response. Expected Response describes the expected behavior(s) of the target device in each test. Prerequisites define the initial conditions that have to be met prior to performing the test. Notes provide informative text relating to the test for context. Details of these four components are described in section 3.2.

The majority of tests are contained in two areas: Section 5: Use Case Test Cases and Section 6: Specific Functionality. Additionally, Section 4 outlines data handling requirements for Test Suite vendors and Section 7 details required test cases for error conditions.

Test cases in Section 5 are required to be performed in sequential order.

Unless otherwise specified, all test cases in this specification apply to all SSCs supported by this specification.

3.2 Test Case Description

3.2.1 Notes

The Notes section is informative; it uses the informative template for informative text: Start of Informative Comment pre-pending statement; End of Informative Comment post-pending statement; and uses of 30% gray background. This component is not populated for every test case.

3.2.2 SSC Applicability

The SSC Applicability section is a section which is used to indicate which SSCs are applicable to the test being performed.

If this section is omitted from a test case, then the test case applies to all SSCs supported by this specification and the test is performed by the Test House for all SSCs, unless otherwise specified.

3.2.3 Prerequisites

Sections 4, 5, and 6 include a set of common prerequisites for each section that need to be met prior to performing any test in that section. Additionally, each test case within a section can have prerequisites specific to that test that need to be met prior to performing the specific test. If there are no prerequisites required for a specific test case, this area states ‘None’ and the test begins with the Test Sequence criteria.

The prerequisites for each test case need to be implemented in sequential order.

3.2.4 Test Sequence

The Test Sequence includes the required steps, in sequential order, that is performed to obtain the Expected Response for a given test. Test Sequences includes different steps for [3] [4] [5] [6] [7] [12] [13] [14] [15] [16].

3.2.5 Expected Response

Expected Response describes the expected behavior(s) of the target device under the Prerequisites and Test Sequence condition(s). All the expected responses are defined in [1] [2] [3] [4] [5] [6] [7] [10] [11] [12] [13] [14] [15] [16] [17].

 © TCG 2025

4 Common Baseline Conditions and Test Criteria

4.1 Minimum Test Requirements

The Test Suite implements:

a) utilize Synchronous Interface Communications capability (See [1]) for host to TPer communications;
b) comply with IF-SEND(s) and IF-RECV(s) command field values described in Table 5;
c) comply with IF-SEND payload field values described in Table 6;
d) contain a payload that does not cause errors or state changes within the TPer (e.g. invocation of the `Properties` method) for tests that require examining the Interface Command Parameters or ComPacket/Packet/Subpacket headers with values other than described above;
e) utilize Read-Write sessions for Regular sessions;
f) adhere to the TPer communications capabilities as reported in the `Properties` method response unless specifically required to do otherwise for a specific test;
g) use the Extended ComID value provided under level 0 Discovery;
h) use the Host Session Number (HSN) 0x00000001, except in the specific Host Session Number (HSN) test defined in section SPF-07: and
i) have a Packet.SeqNumber of 0s for communications sent to the TPer.

The Test Suite does not perform:

a) send empty atoms unless specifically required to do so for a test;
b) utilize Buffer Management capability (See [1]);
c) utilize ACK/NAK capability (See [1]**Error! Reference source not found.**); and
d) trigger any TPer resets unless specifically required to do so for a test.

For invocations of IF-RECV tests, the TPer is in the Awaiting IF-RECV state for a ComID:

a) when the ComPacket header 'OutstandingData' field = 1 the Test Suite re-issues an IF-RECV until the TPer returns a ComPacket header that does not satisfy the condition;
b) when the ComPacket 'OutstandingData' field = <total data available>; and the 'MinTransfer' field = <minimum request length required to transfer a packet>, the Test Suite issues another IF-RECV with greater value of Transfer Length than the previous until the TPer returns a response that does not satisfy the conditions; or
c) when the TPer response contains a Subpacket and the ComPacket 'OutstandingData' field = <additional bytes available, not including the data transferred in the current ComPacket>; and the 'MinTransfer' field = <minimum request required to transfer the next packet>, the Test Suite issues IF-RECV until the TPer returns a response that does not satisfy the above conditions.

Table 5 IF-SEND/RECV Security Protocol=1 Command Field

Field	Value
Security Protocol	1
Security Protocol Specific	any static ComID that the TPer supports and is reported by the Opal SSC Feature Descriptor
Transfer Length	the minimum value necessary to transfer a ComPacket

 © TCG 2025

Table 6 IF-SEND Security Protocol=1 Command Payload

Term	Definition
ComPacket Header	
Reserved	all-0s
ComID	the same value as the Security Protocol Specific field in the IF-SEND
ComID Extension	all-0s
OutstandingData	all-0s
MinTransfer	all-0s
Length	a value that satisfies the following conditions: a) multiple-of-4; b) does not exceed (the TPer's MaxComPacketSize – 20, where 20 is the length of ComPacket header field); and c) indicates its payload contains exactly one Packet.
Packet Header	
Session	a) all-0s for Control session; or b) the session number of the session that was successfully started by a StartSession() and a SyncSession() for Regular session.
SeqNumber	all-0s
Reserved	all-0s
AckType	all-0s
Acknowledgement	all-0s
Length	a value that satisfies the following conditions: a) multiple-of-4; b) does not exceed (the TPer's MaxPacketSize – 24); and c) indicates its payload contains exactly one Subpacket and one Pad field, if necessary.
Subpacket Header	
Reserved	all-0s
Kind	all-0s
Length	a value that is exactly the length of token stream the host is sending to the TPer
Pad	all-0s (and its length is 0 to 3)

 © TCG 2025

4.2 SID Original Factory State Requirement

If Security Identifier (SID) is not Manufacture SID (MSID), the Storage Device (SD) vendor submits the value of SID to Test House and Certification Program Manager (CPM).

4.3 SSC Version Requirement

The SD vendor submits the SSC version implemented by the SD to the Test House and CPM.

Supported SSC versions in this specification include:

1) Opal 1.00 (refer to [3]);
2) Opal 2.00 (refer to [4]);
3) Opal 2.01 (refer to [5]);
4) Opal 2.02 (refer to [6]);
5) Opal 2.30 (refer to [7]);
6) Opalite 1.00 (refer to [12]);
7) Pyrite 1.00 (refer to [13]);
8) Pyrite 2.00 (refer to [14]);
9) Pyrite 2.01 (refer to [15]); or
10) Ruby 1.00 (refer to [16]).

4.4 Feature Set Support Requirement

The SD vendor submits a list of Feature Sets implemented by the SD to the Test House and CPM.

The Test Suite Vendor supports testing against all feature sets supported in this specification.

Supported Feature Sets in this specification include:

1) Additional DataStore Tables, Opal SSC Feature Set (refer to [10]);
2) Physical Presence SID (PSID), Opal SSC Feature Set (refer to [11]); or
3) Block SID Authentication Feature Set (refer to [17]).

4.5 PSID Feature Set Support Requirement

If the SD vendor claims support for the PSID Feature Set, then the SD vendor submits the PSID value to the Test House and CPM.

4.6 Interface Read/Write Command Support Requirement

The SD vendor submits a list of all supported Read and Write commands (as identified by [2]) to the Test House and CPM.

The Test Suite discovers the list of all supported Read and Write commands (as identified by [2]).

 © TCG 2025

5 Use Case Test Cases

5.1 Introduction

Test cases in this section relate to use case scenarios that apply to general SD functionality. Tests in this section are performed in sequential order. Unless otherwise specified within a test case, the expected result of each step is that the step is SUCCEED.

5.2 Common Prerequisites

Unless otherwise noted, the following set of prerequisites apply for each test in this section:

1. Synchronous Protocol state machine for all ComIDs is in “Awaiting IF-SEND” state;
2. The Locking SP is in Manufactured state;
3. The values of any credentials used are known;
4. All `StartSession` method HostChallenge parameters use the current C_PIN object’s PIN column value for the Authority used in the HostSigningAuthority parameter;
5. All sessions are Read-Write sessions; and
6. No open sessions exist at the start of the Test Sequence.

UCT-01: Level 0 Discovery

Notes

Start of Informative Comment

This test includes the sequence of operations required to determine if the SD supports any SSC supported by this specification.

After completing this test, record the ComID for use in later tests.

End of Informative Comment

Prerequisites

None

Test Sequence

1) Issue an IF-RECV Level 0 Discovery with the following conditions:
   a. Security Protocol = 1;
   b. Security Protocol Specific = 0x0001; and
   c. Transfer Length is a value large enough to retrieve the entire response data of Level 0 Discovery.

Expected Response

1) Step #1 SUCCEEDS;
2) The SD returns the following values for Level 0 Discovery:
   a. TPer Feature
      i. Feature Code = 0x0001

 © TCG 2025

ii. Streaming Supported = 1; and
iii. Sync Supported = 1;

and

b. Locking Feature
i. Feature Code = 0x0002;
ii. For Opal 1.00, Opal 2.00, Opal 2.01, Opal 2.02, Opal 2.30, Opalite 1.00, and Ruby 1.00, Media Encryption = 1;
iii. For Pyrite 1.00, Pyrite 2.00 and Pyrite 2.01, Media Encryption = 0; and
iv. Locking Supported = 1;

3) The SD returns the following values for Opal 1.00:
a. Opal 1.00 Feature
i. Feature Code = 0x0200; and
ii. Number of ComIDs >= 1;

4) The SD returns the following values for Opal 2.00, 2.01,2.02 or 2.30:
a. Opal 2.00, 2.01, 2.02 and 2.30 Feature
i. Feature Code = 0x0203;
ii. Number of ComIDs >= 1;
iii. Number of Locking SP Admin Authorities >= 4; and
iv. Number of Locking SP User Authorities >= 8;

and

b. Additional DataStore Table Feature
i. Feature Code = 0x0202;
ii. Maximum number of `DataStore` Tables >= 1;
iii. Maximum total size of `DataStore` Tables >= 0xA0000; and
iv. `DataStore` Table size alignment >= 1;

c. If the device supports Opal v2.02, or 2.30, Block SID Authentication Feature
i. Feature Code = 0x0402; and
ii. SID Value State = 0;

5) The SD returns the following values for Opalite 1.00:
a. Opalite SSC 1.00 Feature
i. Feature Code = 0x0301; and
ii. Number of ComIDs >= 1;

and

b. Block SID Authentication Feature
i. Feature Code = 0x0402; and
ii. SID Value State = 0;

 © TCG 2025

6) The SD returns the following values for Pyrite 1.00:
   a. Pyrite 1.00 Feature
      i. Feature Code = 0x0302; and
      ii. Number of ComIDs >= 1;

      and
   b. Block SID Authentication Feature
      i. Feature Code = 0x0402; and
      ii. SID Value State = 0;
7) The SD returns the following values for Pyrite 2.00 and Pyrite 2.01:
   a. Pyrite 2.00 and Pyrite 2.01Feature
      i. Feature Code = 0x0303; and
      ii. Number of ComIDs >= 1;
   b. Block SID Authentication Feature
      iii. Feature Code = 0x0402; and
      iv. SID Value State = 0;

      and
   c. Supported Data Removal Mechanism Feature
      i. Feature Code = 0x0404;
      ii. Overwrite Data Erase = 1 or Block Erase = 1; and
      iii. Crypto Erase = 0.

and

8) The SD returns the following values for Ruby 1.00:
   a. Ruby 1.00 Feature
      i. Feature Code = 0x0304; and
      ii. Number of ComIDs >= 1;

      and
   b. Block SID Authentication Feature
      i. Feature Code = 0x0402; and
      ii. SID Value State = 0.

UCT-02: Properties

Notes

Start of Informative Comment

The values in the Properties response reported in this section are examples and vary between implementations and locking states of ranges.

End of Informative Comment

 © TCG 2025

Prerequisites

None

Test Sequence

1) Invoke the `Properties` method with the following HostProperties values:
   a. MaxComPacketSize = 4096 bytes;
   b. MaxPacketSize = 4076 bytes; and
   c. MaxIndTokenSize = 4040 bytes.

Expected Response

1) Step #1 SUCCEEDS;
2) The SD returns the following values for TPer Properties:
   a. MaxComPacketSize >= 2048 bytes;
   b. MaxResponseComPacketSize >= 2048 bytes;
   c. MaxPacketSize >= 2028 bytes;
   d. MaxIndTokenSize >= 1992 bytes;
   e. MaxPackets >= 1;
   f. MaxSubpackets >= 1;
   g. MaxMethods >= 1;
   h. MaxSessions >= 1;
   i. MaxAuthentications >= 2;
   j. MaxTranactionLimit >= 1;
   k. DefSessionTimeout >= 0; and
3) The SD returns the following values for Host Properties:
   a. MaxComPacketSize >= 2048 bytes and <= 4096 bytes;
   b. MaxPacketSize >= 2028 bytes and <= 4076 bytes; and
   c. MaxIndTokenSize >= 1992 bytes and <= 4040 bytes.

UCT-03: Taking Ownership of an SD

Notes

Start of Informative Comment

The following test is to establish that an SD can be controlled by host software. Taking ownership is a key step in managing an SD.

End of Informative Comment

 © TCG 2025

Prerequisites

None

Test Sequence

1) If Opal 1.00; or if any other SSC supported by this specification and the Initial C_PIN_SID PIN Indicator value = 0, then:
   a. Invoke the `StartSession` method with SPID = Admin SP UID;
   b. Invoke the `Get` method to retrieve MSID's PIN column value from the `C_PIN` table;
   c. CLOSE_SESSION;
   d. Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
   e. SET_PASSWORD_FOR SID to <SID_PASSWORD>; and
   f. CLOSE_SESSION;
2) If any SSC supported by this specification other than Opal 1.00 and the Initial C_PIN_SID PIN Indicator value <> 0, then obtain SID VU PIN value from the SD vendor
   a. Invoke the `StartSession` method with SPID = Admin SP UID, HostSigningAuthority = SID authority UID, and HostChallenge = SID C_PIN object's VU PIN column value/
   b. SET_PASSWORD_FOR SID to <SID_PASSWORD> and
   c. CLOSE_SESSION;

   and
3) If Opal 2.00, 2.01, 2.02 or 2.30
   a. Invoke the `StartSession` method with SPID = Admin SP UID, HostSigningAuthority = SID authority UID, and HostChallenge = <SID_PASSWORD>;
   b. SET_PASSWORD_FOR Admin1 to <AdminSP_Admin1_ PASSWORD>;
   c. ENABLE Admin1; and
   d. CLOSE_SESSION.

Expected Response

1) If Opal 1.00, or if any other SSC supported by this specification and the Initial C_PIN_SID PIN Indicator value = 0, then step #1 SUCCEEDS;
2) If any SSC supported by this specification other than Opal 1.00 and the Initial C_PIN_SID PIN Indicator value <> 0, then step #2 SUCCEEDS; and
3) If Opal 2.00, 2.01, 2.02 or 2.30 then step #3 SUCCEEDS.

 © TCG 2025

UCT-04: Activate Locking SP when in Manufactured-Inactive State

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

1) Locking SP is in the Manufactured-Inactive state; and

Test Sequence

1) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
2) Invoke the `Activate` method on Locking SP object;
3) CLOSE_SESSION;
4) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID; and
5) CLOSE_SESSION.

Expected Response

1) Steps #1-5 SUCCEED

UCT-05: Configuring Authorities

Notes

Start of Informative Comment

The following sections describe the sequences of steps for setting the PIN Credential value for one or more Admin authorities, and enabling and setting the PIN Credential value for multiple User authorities.

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) SET_PASSWORD_FOR Admin1 to <Admin1_PASSWORD>;
3) ENABLE *User1;*
4) SET_PASSWORD_FOR *User1 to <User1_PASSWORD>;*

 © TCG 2025

5) Enable LAST_REQUIRED_USER;
6) SET_PASSWORD_FOR LAST_REQUIRED_USER to <LAST_REQUIRED_USER_PASSWORD>
7) CLOSE_SESSION;
8) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
9) CLOSE_SESSION;
10) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = User1 authority UID;
11) CLOSE_SESSION;
12) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = LAST_REQUIRED_USER authority UID; and
13) CLOSE_SESSION.

Expected Response

1) Steps #1-13 SUCCEED

UCT-06: Configuring Locking Objects (Locking Ranges)

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID
2) This test step varies based on the SSC version:
   a. For Opal, invoke the `Set` method on LAST_REQUIRED_RANGE. Configure the locking range as follows:
      i. RangeStart = 0;
      ii. RangeLength = 64;
      iii. ReadLockEnabled = TRUE;
      iv. WriteLockEnabled = TRUE;
      v. ReadLocked = FALSE;
      vi. WriteLocked = FALSE;
      vii. For Opal 2.00, 2.01, 2.02 and 2.30, LockOnReset**=** {0}; and

 © TCG 2025

viii. Adjust RangeStart and RangeLength according to the Range Start Alignment and Range Length Alignment;

b. For all SSCs supported by this specification other than Opal, invoke the `Set` method on Locking_GlobalRange. Configure the locking range as follows:

i. ReadLockEnabled = TRUE;

ii. WriteLockEnabled = TRUE;

iii. ReadLocked = FALSE;

iv. WriteLocked = FALSE; and

v. LockOnReset **=** {0};

3) Invoke the `Set` method on the BooleanExpr column of the LAST_REQUIRED_RANGE_RDLOCKED_ACE ACE object to set the UIDs of the User1 and LAST_REQUIRED_USER Authority objects;

4) Invoke the `Set` method on the BooleanExpr column of the LAST_REQUIRED_RANGE_WRLOCKED_ACE ACE object to set the UIDs of the User1 and LAST_REQUIRED_USER Authority objects***;***

5) CLOSE_SESSION;

6) This test step varies based on the SSC version:

   a. For Opal, Write the MAGIC_PATTERN over the entire LAST_REQUIRED_RANGE; and

   b. For all SSCs supported by this specification other than Opal, Write the MAGIC_PATTERN over an ARBITRARILY_VARYING_LBA_RANGE;

7) This test step varies based on the SSC version:

   a. For Opal, Read the entire LAST_REQUIRED_RANGE; and

   b. For all SSCs supported by this specification other than Opal, Read the same ARBITRARILY_VARYING_LBA_RANGE in Step #6;

8) Power cycle the SD;

9) Invoke the Get method on the Locking_GlobalRange ReadLocked, WriteLocked, ReadLockEnabled, WriteLockEnabled columns;

10) This test step varies based on the SSC version:

    a. For Opal, Read the entire LAST_REQUIRED_RANGE; and

    b. For all SSCs supported by this specification other than Opal, Read the same ARBITRARILY_VARYING_LBA_RANGE in Step #6;

    and

11) This test step varies based on the SSC version:

    a. For Opal, Write the MAGIC_PATTERN over the entire LAST_REQUIRED_RANGE; and

    b. For all SSCs supported by this specification other than Opal, Write the MAGIC_PATTERN over an ARBITRARILY_VARYING_LBA_RANGE.

Expected Response

1) Steps #1-8 SUCCEED;

2) The value returned from the Read command in step #7 is the MAGIC_PATTERN;

 © TCG 2025

3) The Get method in step #9 returns values of TRUE; and
4) Steps #10-11 return an interface status of Data Protection Error, see [2].

UCT-07: Unlocking Ranges

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = User1 authority UID;
2) Invoke the `Set` method on the ReadLocked and WriteLocked columns of the LAST_REQUIRED_RANGE Locking object with a value of FALSE;
3) CLOSE_SESSION; and
4) This test step varies based on the SSC version:
   a. For Opal, Read the entire LAST_REQUIRED_RANGE; and
   b. For all SSCs supported by this specification other than Opal, Read an ARBITRARILY_VARYING_LBA_RANGE.

Expected Results

1) Steps #1-4 SUCCEED

UCT-08: Erasing Ranges

Notes

Start of Informative Comment

None

End of Informative Comment

SSC Applicability

This test case applies to all SSCs supported by this specification except for:

1) Pyrite 1.00;

 © TCG 2025

2) Pyrite 2.00; and
3) Pyrite 2.01.

Prerequisites

None

Test Sequence

1) This test step varies based on the SSC version:
   a. For Opal, Write the MAGIC_PATTERN over the entire LAST_REQUIRED_RANGE; and
   b. For all SSCs supported by this specification other than Opal, Write the MAGIC_PATTERN over an ARBITRARILY_VARYING_LBA_RANGE;
2) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
3) Invoke the `Get` method on the LAST_REQUIRED_RANGE to retrieve the ActiveKey column's value;
4) Invoke the `GenKey` method on the UID retrieved from the LAST_REQUIRED_RANGE's ActiveKey column;
5) CLOSE_SESSION; and
6) This test step varies based on the SSC version:
   a. For Opal, attempt to read the entire LAST_REQUIRED_RANGE; and
   b. For all SSCs supported by this specification other than Opal, attempt to read the entire ARBITRARILY_VARYING_LBA_RANGE that was written to in test step #1.

Expected Response

1) Steps #1-5 SUCCEED; and
2) The Read command in step #6 responds in one of the following ways:
   a. The Read command FAILS and no data is transferred;
   b. The Read command FAILS, data is transferred, and the returned data does not match the MAGIC_PATTERN; or
   c. The Read command SUCCEEDS, data is transferred, and the returned data does not match the MAGIC_PATTERN.

UCT-09: Using the DataStore Table

Notes

Start of Informative Comment

None

End of Informative Comment

 © TCG 2025

Prerequisites

The size of the DataStore byte table is not zero

Test Sequence

1) Invoke the StartSession method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Invoke the Set method on the BooleanExpr column of the ACE_DataStore_Set_All ACE object to include the UID of the User1 Authority object;
3) Invoke the Set method on the BooleanExpr column of the ACE_DataStore_Get_All ACE object to include the UID of the User1 Authority object;
4) CLOSE_SESSION;
5) Invoke the StartSession method with SPID = Locking SP UID and HostSigningAuthority = User1 authority UID;
6) Invoke the Set method to write the entire DataStore table with the MAGIC_PATTERN;
7) CLOSE_SESSION;
8) Invoke the StartSession method with SPID = Locking SP UID and HostSigningAuthority = User1 authority UID;
9) Invoke the Get method on the DataStore table to read the data of the DataStore table; and
10) CLOSE_SESSION.

Expected Response

1) Steps #1-10 SUCCEED; and
2) The Get method in step #9 returns the MAGIC_PATTERN.

UCT-10: Enable MBR Shadowing

Notes

Start of Informative Comment

None

End of Informative Comment

SSC Applicability

This test case applies to all SSCs supported by this specification with the following exception for Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00:

This test case only applies to Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00 if the MBR Shadowing feature is supported.

 © TCG 2025

Prerequisites

The size of the MBR table is not zero

Test Sequence

1) Invoke the StartSession method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Invoke the Set method on the BooleanExpr column of the ACE_MBRCONTROL_SET_DONE ACE object to include the UIDs of the User1 and LAST_REQUIRED_USER Authority objects;
3) Invoke the Get method on the Rows column of the MBR Table Descriptor Object;
4) This test step varies based on the SSC version:
   a. For Opal, invoke the Set method to change the RangeLength column of the LAST_REQUIRED_RANGE to SIZE_OF_MBR_TABLE_DESCRIPTOR_IN_LOGICAL_BLOCKS + 10 LBAs and adjust RangeLength according to the Range Length Alignment; and
   b. For Opalite 1.00, Pyrite 1.00, Pyrite 2.00, Pyrite 2.01, and Ruby 1.00, Do nothing for this step;
5) This test step varies based on the SSC version:
   a. For Opal, write 1s over the entire LAST_REQUIRED_RANGE; and
   b. For all SSCs supported by this specification other than Opal, write 1s over the range from LBA 0 to SIZE_OF_MBR_TABLE_DESCRIPTOR_IN_LOGICAL_BLOCKS + 10;
6) This test step varies based on the SSC version:
   a. For Opal 1.00 invoke the Set method to write the entire MBR table with the MAGIC_PATTERN; and
   b. For all SSCs supported by this specification other than Opal 1.00, invoke the Set method to write the entire MBR table with the MAGIC_PATTERN while adhering to the MandatoryWriteGranularity requirements;
7) Invoke the Set method on the Enable column of the MBRControl table with a value of TRUE;
8) CLOSE_SESSION;
9) Power cycle the SD;
10) Invoke the Get method on the Locking_GlobalRange ReadLocked, WriteLocked, ReadLockEnabled, WriteLockEnabled columns;
11) This test step varies based on the SSC version:
    a. For Opal, Write the MAGIC_PATTERN over the entire LAST_REQUIRED_RANGE; and
    b. For all SSCs supported by this specification other than Opal, Write the MAGIC_PATTERN over the entire range from LBA 0 to SIZE_OF_MBR_TABLE_DESCRIPTOR_IN_LOGICAL_BLOCKS + 10;
12) Read from LBA 0 to the size of the MBR Table;

    and

13) This test step varies based on the SSC version:
    a. For Opal 1.00 Read 10 LBAs starting immediately following the end of the MBR; and
    b. For all SSCs supported by this specification other than Opal 1.00, Read 10 LBAs or an appropriate value adhering to the Range Start Alignment requirements, starting immediately following the end of the MBR Shadowing.

 © TCG 2025

Expected Response

1) Steps #1-9 SUCCEED;
2) The Get method in step #10 returns values of TRUE;
3) Step #11 returns Data Protection Error;
4) The value returned from the Read command in step #12 matches the MAGIC_PATTERN written in step #6; and

Notes

Start of Informative Comment

The locking range was locked due to the power cycle issued on #9 and the ReadLockedEn in test UCT-06 and ReadLocked was set to True after step#9. So when the MBR shadow is enabled and a range is read locked, then attempting to read the range (if it's not withing the MBR shadow region) will result in 0s being returned. See [1].

End of Informative Comment

5) The value returned from the Read command in step #13 = 0s.

UCT-11: MBR Done

Notes

Start of Informative Comment

None

End of Informative Comment

SSC Applicability

This test case applies to all SSCs supported by this specification with the following exception for Pyrite 1.00, Pyrite 2.00, Pyrite 2.01, and Ruby 1.00:

This test case only applies to Pyrite 1.00, 2.00, 2.01 and Ruby 1.00 if the MBR Shadowing feature is supported.

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = LAST_REQUIRED_USER authority UID;
2) Invoke the `Set` method on the ReadLocked and WriteLocked columns of the LAST_REQUIRED_RANGE Locking object with a value of FALSE;
3) Invoke the `Set` method on the Done column of the `MBRControl` table with a value of TRUE;

 © TCG 2025

4) CLOSE_SESSION; and
5) This test step varies based on SSC version:
   a. For Opal, read the entire LAST_REQUIRED_RANGE; and
   b. For all SSCs supported by this specification other than Opal, read the entire range from LBA 0 to SIZE_OF_MBR_TABLE_DESCRIPTOR_IN_LOGICAL_BLOCKS + 10.

Expected Response

1) Steps #1-5 SUCCEED; and
2) The value returned from the Read command in step #5 = 1s.

UCT-12: Revert the Locking SP using SID, with Locking SP in Manufactured state

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

For ZNS device, reset Write Pointer

Test Sequence

1) Write the MAGIC_PATTERN over 64 logical blocks beginning at LBA 0;
2) For ZNS device, read Write Pointer and Zone State;
3) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
4) Invoke the `Revert` method on Locking SP object;
5) CLOSE_SESSION;
6) Invoke the StartSession method with SPID = Locking SP UID and HostSigningAuthority = Anybody authority UID;
7) This test step varies based on the SSC version:
   a. For all SSCs supported by this specification other than Pyrite 1.00, read 64 logical blocks beginning at LBA 0; and
   b. For Pyrite 1.00, do nothing for this step;
   and
8) For ZNS device, read Write Pointer and Zone State.

Expected Response

1) Steps #1-5 SUCCEED;

 © TCG 2025

2) The `StartSession` method in step #6 results in a `SyncSession` method with a status code of INVALID_PARAMETER;
3) For all SSCs supported by this specification other than Pyrite 1.00, the Read command in step #6 responds in one of the following ways:
   a. The Read command FAILS and no data is transferred;
   b. The Read command FAILS, data is transferred, and the returned data does not match the MAGIC_PATTERN; or
   c. The Read command SUCCEEDS, data is transferred, and the returned data does not match the MAGIC_PATTERN;

   and
4) For ZNS device in step #8,
   a. If Key Change Zone Behavior bit of the SIIS Feature Descriptor is equal to one, then the Write Pointer = 0 and Zone State = Empty; or
   b. If Key Change Zone Behavior bit of the SIIS Feature Descriptor is equal to zero, then the Write Pointer and Zone State keep same as the ones read in step #2.

UCT-13: Revert the Admin SP using SID, with Locking SP in Manufactured-Inactive state

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

For ZNS device, reset Write Pointer

Test Sequence

1) Write the MAGIC_PATTERN over 64 logical blocks beginning at LBA 0;
2) For ZNS device, read Write Pointer and Zone State;
3) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
4) Invoke the `Revert` method on Admin SP object;
5) If the “Behavior of C_PIN_SID Pin upon TPer Revert” from the return of Level 0 Discovery = 0 then

   Invoke the `StartSession` method with SPID = Admin SP UID, HostSigningAuthority = SID authority UID, and HostChallenge = C_PIN_MSID PIN column value

   Else

   Invoke the `StartSession` method with SPID = Admin SP UID, HostSigningAuthority = SID authority UID, and HostChallenge = C_PIN_SID VU PIN column value;

 © TCG 2025

6) CLOSE_SESSION;
7) Invoke the StartSession method with SPID = Locking SP UID and HostSigningAuthority = Anybody authority UID;
8) Read 64 logical blocks beginning at LBA 0; and
9) For ZNS device, read Write Pointer and Zone State.

Expected Response

1) Steps #1-6 SUCCEED;
2) The `StartSession` method in step #7 results in a `SyncSession` method with a status code of INVALID_PARAMETER;
3) The Read command in step #8 returns data that matches the MAGIC_PATTERN; and
4) For ZNS device in step #9, Write Pointer and Zone State is the same as those read in step #2.

UCT-14: Revert the Admin SP using SID, with Locking SP in Manufactured state

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

1) SID's PIN column value is set to <SID_PASSWORD> value in the SID's C_PIN credential PIN column;
2) Locking SP is in the Manufactured state; and
3) For ZNS device, reset Write Pointer.

Test Sequence

1) Write the MAGIC_PATTERN over 64 logical blocks beginning at LBA 0;
2) For ZNS device, read Write Pointer and Zone State;
3) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
4) Invoke the `Revert` method on Admin SP object;
5) If the "Behavior of C_PIN_SID Pin upon TPer Revert" from the return of Level 0 Discovery = 0 then

   Invoke the `StartSession` method with SPID = Admin SP UID, HostSigningAuthority = SID authority UID, and HostChallenge = C_PIN_MSID PIN column value

   Else

   Invoke the `StartSession` method with SPID = Admin SP UID, HostSigningAuthority = SID authority UID, and HostChallenge = C_PIN_SID VU PIN column value;
6) CLOSE_SESSION;

 © TCG 2025

7) Invoke the StartSession method with SPID = Locking SP UID and HostSigningAuthority = Anybody authority UID;
8) Read 64 logical blocks beginning at LBA 0; and
9) For ZNS device, read Write Pointer and Zone State.

Expected Response

1) Steps #1-6 SUCCEED;
2) The `StartSession` method in step #7 results in a `SyncSession` method with a status code of INVALID_PARAMETER;
3) For all SSCs supported by this specification other than Pyrite 1.00, The Read command in step #8 responds in one of the following ways:
   a. The Read command FAILS and no data transfer;
   b. The Read command FAILS and there is data transfer. And the return data does not match the MAGIC_PATTERN; or
   c. The Read command SUCCEEDS and there is data transfer. And the return data does not match the MAGIC_PATTERN;

   and
4) For ZNS device in step #9,
   a. If the Key Change Zone Behavior bit of the SIIS Feature Descriptor is equal to one, then the Write Pointer = 0 and Zone State = Empty; or
   b. If the Key Change Zone Behavior bit of the SIIS Feature Descriptor is equal to zero, then the Write Pointer and Zone State keep the same as those read in step #2.

UCT-15: Revert Admin SP using Admin1, with Locking SP in Manufactured state

Notes

Start of Informative Comment

See [4] [5] [6] [7] [12] [13] [14] [15] [16] for support requirements.

End of Informative Comment

SSC Applicability

This test case applies to Opal 2.00, 2.01, 2.02 and 2.30 with no exceptions.

This test case only applies to all other SSCs supported by this specification if the Admin1 authority in the `Authority` table of the AdminSP is implemented.

Prerequisites

1) Locking SP is in the Manufactured state;
2) Admin1 in the Admin SP authority is enabled

 © TCG 2025

Admin1's PIN column value is set to <Admin1_PASSWORD> value in the Admin1's C_PIN credential PIN column; and

3) For ZNS device, reset Write Pointer.

Test Sequence

1) Write the MAGIC_PATTERN over 64 logical blocks beginning at LBA 0;
2) For ZNS device, read Write Pointer and Zone State;
3) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = Admin1 authority UID;
4) Invoke the `Revert` method on Admin SP object;
5) If the "Behavior of C_PIN_SID Pin upon TPer Revert" from the return of Level 0 Discovery = 0 then

   Invoke the `StartSession` method with SPID = Admin SP UID, HostSigningAuthority = SID authority UID, and HostChallenge = C_PIN_MSID PIN column value

   Else

   Invoke the `StartSession` method with SPID = Admin SP UID, HostSigningAuthority = SID authority UID, and HostChallenge = C_PIN_SID VU PIN column value;
6) CLOSE_SESSION;
7) Invoke the `StartSession` method with SPID = Locking SP UID;
8) Read 64 logical blocks beginning at LBA 0; and
9) For ZNS device, read Write Pointer and Zone State.

Expected Response

1) Steps #1-6 SUCCEED;
2) The `StartSession` method in step #7 results in a `SyncSession` method with a status code of INVALID_PARAMETER;
3) For all SSCs supported by this specification other than Pyrite 1.00, the Read command in step #8 responds in one of the following ways:
   a. The Read command FAILS and no data is transferred;
   b. The Read command FAILS, data is transferred, and the returned data that does not match the MAGIC_PATTERN; or
   c. The Read command SUCCEEDS, data is transferred, and the returned data does not match the MAGIC_PATTERN;

   and
4) For ZNS device in step #9,
   a. If Key Change Zone Behavior bit of the SIIS Feature Descriptor is equal to one, then the Write Pointer = 0 and Zone State = Empty; or
   b. If Key Change Zone Behavior bit of the SIIS Feature Descriptor is equal to zero, then the Write Pointer and Zone State keep the same as the one read in step #2.

 © TCG 2025

UCT-16: Revert Admin SP using PSID, with Locking SP in Manufactured state

Notes

Start of Informative Comment

None

End of Informative Comment

SSC Applicability

This test case applies to the following SSCs:

1) Opal 2.01;
2) Opal 2.02;
3) Opal 2.30;
4) Opalite 1.00;
5) Pyrite 2.00;
6) Pyrite 2.01; or
7) Ruby 1.00.

If the PSID Feature Set is implemented this test case also applies to Opal 1.00 and Opal 2.00.

Prerequisites

1) The PIN column of C_PIN_SID is set to <SID_PASSWORD>;
2) Locking SP is in the Manufactured state; and
3) For ZNS device, reset Write Pointer.

Test Sequence

1) Write the MAGIC_PATTERN over 64 logical blocks beginning at LBA 0;
2) For ZNS device, read Write Pointer and Zone State;
3) Invoke the `StartSession` method with SPID = Admin SP UID, HostSigningAuthority = PSID authority UID, and HostChallenge = PSID authority's credential obtained from the VU PSID delivery mechanism;
4) Invoke the `Revert` method on Admin SP object;
5) If the "Behavior of C_PIN_SID Pin upon TPer Revert" from the return of Level 0 Discovery = 0 then

   Invoke the `StartSession` method with SPID = Admin SP UID, HostSigningAuthority = SID authority UID, and HostChallenge = C_PIN_MSID PIN column value

   Else

   Invoke the `StartSession` method with SPID = Admin SP UID, HostSigningAuthority = SID authority UID, and HostChallenge = C_PIN_SID VU PIN column value;
6) CLOSE_SESSION;

 © TCG 2025

7) Invoke the `StartSession` method with SPID = Locking SP UID;
8) Read 64 logical blocks beginning at LBA 0; and
9) For ZNS device, read Write Pointer and Zone State.

Expected Response

1) Steps #1-6 SUCCEED;
2) The `StartSession` method in step #7 results in a `SyncSession` method with a status code of INVALID_PARAMETER;
3) The Read command in step #8 responds in one of the following ways:
   a. The Read command FAILS and no data is transferred;
   b. The Read command FAILS, data is transferred, and the returned data does not match the MAGIC_PATTERN; or
   c. The Read command SUCCEEDS, data is transferred, and the returned data does not match the MAGIC_PATTERN;

   and
4) For ZNS device in step #9,
   a. If Key Change Zone Behavior bit of the SIIS Feature Descriptor is equal to one, then the Write Pointer = 0 and Zone State = Empty
   b. If Key Change Zone Behavior bit of the SIIS Feature Descriptor is equal to zero, then the Write Pointer and Zone State keep the same as those read in step #2.

 © TCG 2025

6 Specific Functionality

6.1 Introduction

These test cases reflect specific functionality that is performed on a device that complies with the Opal 1.00, Opal 2.00, 2.01, 2.02 and 2.30, Opalite 1.00, Pyrite 1.00, Pyrite 2.00, Pyrite 2.01, or Ruby 1.00 specifications. Unless otherwise specified within a test case, the expected result of each step is that the step is SUCCEED.

6.2 Common Prerequisites

Unless otherwise noted, the following set of prerequisites apply for each test in this section:

1. SD is in Awaiting IF-SEND;
2. Locking SP is in Manufactured state;
3. The values of any credentials used are known;
4. All `StartSession` method HostChallenge parameters use the current C_PIN object's PIN column value for the Authority used in the HostSigningAuthority parameter;
5. All sessions are Read-Write sessions; and
6. No open sessions exist at the start of the Test Sequence.

SPF-01: Transaction

Notes

Start of Informative Comment

There are two tests performed relating to Transactions:

1. Case 1 attempts to write an entire table with the MAGIC_PATTERN.

2. Case 2 attempts to write an entire table with 0s, and then close the session without committing the Transaction.

In most cases, the `MBR` table is used for these tests but for SSCs where the MBR shadowing feature is optional, the `MBR` table is only used when the MBR shadowing feature is supported, otherwise the `DataStore` table is used.

Since Session Timeout is VU, test results is NA if session timeout occurs during writing of the Shadow MBR Table or if the transaction cannot be committed.

End of Informative Comment

Case 1:

Prerequisites

1) For Opal 1.00, 2.00, 2.01, 2.02 and 2.30, and Opalite 1.00, knowledge of the `MBR` table size;
2) For Opal 2.00, 2.01, 2.02 and 2.30, and Opalite 1.00, knowledge of the MandatoryWriteGranularity Column value for the `MBR` table;
3) For Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00, if the MBR Shadowing feature is supported, then knowledge of the `MBR` table size, otherwise knowledge of the `DataStore` table size;
4) For Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00, if the MBR Shadowing feature is supported, then knowledge of the MandatoryWriteGranularity Column value for the `MBR` table, otherwise knowledge of the MandatoryWriteGranularity Column value for the `DataStore` table; and

 © TCG 2025

5) The size of the MBR table is not zero.

Test Sequence

1) Invoke the StartSession method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) This test step varies based on SSC version:
   a. For Opal 1.00, invoke the Set method to write the entire MBR table with 0s;
   b. For Opal 2.00, 2.01, 2.02 and 2.30, and Opalite 1.00, invoke the Set method to write the entire MBR table with 0s while adhering to the MandatoryWriteGranularity requirements;
   c. For Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00, if the MBR Shadowing feature is supported, invoke the Set method to write the entire MBR table with 0s while adhering to the MandatoryWriteGranularity requirements; and
   d. For Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00, if the MBR Shadowing Feature is not supported, invoke the Set method to write the entire DataStore table with 0s while adhering to the MandatoryWriteGranularity requirements;
3) CLOSE_SESSION if the write is successful, or if the session aborts due to a timeout, exit the test and record result as NA;
4) Invoke the StartSession method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
5) Send a subpacket that contains a StartTransaction token with a status code of 0x00;
6) This test step varies based on SSC version:
   a. For Opal 1.00, invoke the Set method to write the entire MBR table with the MAGIC_PATTERN;
   b. For Opal 2.00, 2.01, 2.02 and 2.30, and Opalite 1.00, invoke the Set method to write the entire MBR table with the MAGIC_PATTERN while adhering to the MandatoryWriteGranularity requirements;
   c. For Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00, if the MBR Shadowing feature is supported, invoke the Set method to write the entire MBR table with the MAGIC_PATTERN while adhering to the MandatoryWriteGranularity requirements; and
   d. For Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00, if the MBR Shadowing feature is not supported, invoke the Set method to write the entire DataStore table with the MAGIC_PATTERN while adhering to the MandatoryWriteGranularity requirements;
7) Send a subpacket that contains an End Transaction token with a status code of 0x00;
8) CLOSE_SESSION if the SD responds with an End Transaction token with a status code of 0x00, or if the session aborts due to a timeout exit the test and record result as NA;
9) Invoke the StartSession method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
10) This test step varies based on SSC version:
   a. For Opal 1.00, 2.00, 2.01, 2.02 and 2.30, and Opalite1.00, invoke the Get method on the MBR table to read the data from the table;
   b. For Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00, if the MBR Shadowing feature is supported, invoke the Get method on the MBR table to read the data from the table; and

 © TCG 2025

c. For Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00, if the MBR Shadowing feature is not supported, invoke the `Get` method on the `DataStore` table to read the data from the table;

and

11) CLOSE_SESSION.

Expected Response

1) Steps #1-11 SUCCEED;
2) The `Get` in step #10 returns the MAGIC_PATTERN; and
3) If the session is aborted on step #3 or step #8, the result of this test is NA.

Case 2:

Prerequisites

1) Steps #1-11 in Case 1 SUCCEED

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Send a subpacket that contains a StartTransaction token with a status code of 0x00;
3) This test step varies based on SSC version:
   a. For Opal 1.00, invoke the `Set` method to write the entire `MBR` table with 0s;
   b. For Opal 2.00, 2.01, 2.02 and 2.30, and Opalite 1.00, invoke the `Set` method to write the entire `MBR` table with 0s while adhering to the MandatoryWriteGranularity requirements;
   c. For Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00, if the MBR Shadowing feature is supported, invoke the `Set` method to write the entire `MBR` table with 0s while adhering to the MandatoryWriteGranularity requirements; and
   d. For Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00, if the MBR Shadowing feature is not supported, invoke the `Set` method to write the entire `DataStore` table with 0s while adhering to the MandatoryWriteGranularity requirements;
4) CLOSE_SESSION if the write is successful, or if the session aborts due to a timeout exit the test and record result as NA;
5) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
6) This test step varies based on SSC version:
   a. For Opal 1.00, 2.00, 2.01, 2.02 and 2.30, and Opalite 1.00, invoke the `Get` method on the `MBR` table to read the data from the table;
   b. For Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00, if the MBR Shadowing feature is supported, invoke the `Get` method on the `MBR` table to read the data from the table; and

 © TCG 2025

c. For Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00, if the MBR Shadowing feature is not supported, invoke the `Get` method on the `DataStore` table to read the data from the table;

and

7) CLOSE_SESSION.

Expected Response

1) Steps #1-7 SUCCEED;
2) The `Get` method in step #6 returns the MAGIC_PATTERN in the case 1 step #6; and
3) If the session is aborted on step #4, the result of this test is NA.

SPF-02: IF-RECV Behavior Tests

Notes

Start of Informative Comment

There are two tests performed relating to IF-RECV Behavior:

Case 1 attempts to issue an IF-RECV command while the SD is in an Awaiting IF-SEND state

Case 2 attempts to issue an IF-RECV command with an Insufficient Transfer Length

End of Informative Comment

Case 1:

Prerequisites

1) In Awaiting IF-SEND

Test Sequence

1) Issue an IF-RECV command

Expected Response

1) Steps #1 SUCCEEDS; and
2) IF-RECV in step #1 has a ComPacket header value of “All Response(s) returned - no further data”, (See [1]).

Case 2:

Prerequisites

None

 © TCG 2025

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Invoke the `Get` method on the `DataStore` table to retrieve 1024 Rows. For the IF-RECV command issued by the Host to retrieve the result, the IF-RECV command has a transfer length of 1;
3) Issue IF-RECV command to retrieve the result with the transfer length based on the MinTransfer value in the IF-RECV response to step #2; and
4) CLOSE_SESSION.

Expected Response

1) Step #1-4 SUCCEED; and
2) IF-RECV in step #2 has a ComPacket header value of “Response ready, insufficient transfer length request”, see [1].

SPF-03: TryLimit

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

1) User1 is enabled

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Invoke the `Get` method on Admin1’s C_PIN Object to retrieve the TryLimit Column’s value;
3) Invoke the `Get` method on User1’s C_PIN Object to retrieve the TryLimit Column’s value;
4) CLOSE_SESSION;
5) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
6) Invoke the `Get` method on SID’s C_PIN Object to retrieve the TryLimit Column’s value;
7) CLOSE_SESSION;
8) If SID C_PIN Object has a TryLimit Column value >0, then
    a. Invoke the `StartSession` method with SPID = Admin SP UID, HostSigningAuthority = SID authority UID, and HostChallenge = a value that does not match the current SID C_PIN object’s PIN column value, until SID C_PIN object’s Tries value = SID C_PIN object’s TryLimit value; and

 © TCG 2025

b. Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID

Else do not perform this test step and the test step is NA;

9) If Admin1 C_PIN Object has a TryLimit Column value >0, then

a. Invoke the `StartSession` method with SPID = Locking SP UID, HostSigningAuthority = Admin1 authority UID, and HostChallenge = a value that does not match the current Admin1 C_PIN object's PIN column value, until Admin1 C_PIN object's Tries value = Admin1 C_PIN object's TryLimit value; and

b. Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID

Else do not perform this test step and the Test Suite marks the result of this step as NA;

and

10) If User1 C_PIN Object has a TryLimit Column value >0, then

a. Invoke the `StartSession` method with SPID = Locking SP UID, HostSigningAuthority = User1 authority UID, and HostChallenge = a value that does not match the current User1 C_PIN object's PIN column value, until User1 C_PIN object's Tries value = User1 C_PIN object's TryLimit value; and

b. Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = User1 authority UID

Else do not perform this test step and the Test Suite marks the result of this step as NA.

Expected Response

1) Steps #1-7 SUCCEED;
2) Steps #8-10 FAIL for any Authority with a TryLimit value >0;
3) Every `StartSession` method in steps #8a, #9a, and #10a results in a `SyncSession` method with a status code of NOT_AUTHORIZED; and
4) The `StartSession` method with the correct HostChallenge value in steps #8b, #9b, and #10b results in a `SyncSession` method with a status code of AUTHORITY_LOCKED_OUT.

SPF-04: Tries Reset

Notes

Start of Informative Comment

The following test verifies that the value of Tries is reset upon successful authentication.

End of Informative Comment

Prerequisites

1) User1 is enabled

 © TCG 2025

Test Sequence

1) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
2) Invoke the `Get` method on SID's C_PIN Object to retrieve the TryLimit Column's value;
3) CLOSE_SESSION;
4) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
5) Invoke the `Get` method on Admin1's C_PIN Object to retrieve the TryLimit Column's value;
6) Invoke the `Get` method on User1's C_PIN Object to retrieve the TryLimit Column's value;
7) CLOSE_SESSION;
8) If SID C_PIN Object has a TryLimit Column value > 1, then
    a. Invoke the `StartSession` method with SPID = Admin SP UID, HostSigningAuthority = SID authority UID, and HostChallenge = a value that does not match the current SID C_PIN object's PIN column value, until SID C_PIN object's Tries value = SID C_PIN object's TryLimit value -1;
    b. Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
    c. Invoke the `Get` method on the Tries Column of the SID Authority's C_PIN Object; and
    d. CLOSE_SESSION;
9) If Admin1 C_PIN Object has a TryLimit Column value > 1, then
    a. Invoke the `StartSession` method with SPID = Locking SP UID, HostSigningAuthority = Admin1 authority UID, and HostChallenge = a value that does not match the current Admin1 C_PIN object's PIN column value, until Admin1 C_PIN object's Tries value = Admin1 C_PIN object's TryLimit value -1;
    b. Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
    c. Invoke the `Get` method on the Tries Column of the Admin1 Authority's C_PIN Object; and
    d. CLOSE_SESSION;

    and
10) If User1 C_PIN Object has a TryLimit Column value >1, then
    a. Invoke the `StartSession` method with SPID = Locking SP UID, HostSigningAuthority = User1 authority UID, and HostChallenge = a value that does not match the current User1 C_PIN object's PIN column value, until User1 C_PIN object's Tries value = User1 C_PIN object's TryLimit value -1;
    b. Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = User1 authority UID;
    c. CLOSE_SESSION;
    d. Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
    e. Invoke the `Get` method on the Tries Column of the User1 Authority's C_PIN Object; and
    f. CLOSE_SESSION.

 © TCG 2025

Expected Response

1) Steps #1-10 SUCCEED; and
2) For each Authority with a TryLimit column value > 1, that Authority's C_PIN Tries column value = 0 on steps #8c, #9c, and #10e.

SPF-05: Tries Reset on Power Cycle

Notes

Start of Informative Comment

The following test verifies that the value of Tries is reset upon power cycle.

End of Informative Comment

Prerequisites

1) User1 is enabled

Test Sequence

1) If the Persistence Column of C_PIN object for all authorities (SID/Admin1/User1 to be tested) = FALSE, continue: if the Persistence Column of C_PIN object for any authority (SID/Admin1/User1 to be tested) = TRUE, exit the test sequence;
2) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
3) Invoke the `Get` method on SID's C_PIN Object to retrieve the TryLimit Column's value;
4) CLOSE_SESSION;
5) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
6) Invoke the `Get` method on Admin1's C_PIN Object to retrieve the TryLimit Column's value;
7) Invoke the `Get` method on User1's C_PIN Object to retrieve the TryLimit Column's value;
8) CLOSE_SESSION;
9) If SID C_PIN Object has a TryLimit Column value >0, then

   Invoke the `StartSession` method with SPID = Admin SP UID, HostSigningAuthority = SID authority UID, and HostChallenge = a value that does not match the current SID C_PIN object's PIN column value, until SID C_PIN object's Tries value = SID C_PIN object's TryLimit value;
10) If Admin1 C_PIN Object has a TryLimit Column value >0, then

    Invoke the `StartSession` method with SPID = Locking SP UID, HostSigningAuthority = Admin1 authority UID, and HostChallenge = a value that does not match the current Admin1 C_PIN object's PIN column value, until Admin1 C_PIN object's Tries value = Admin1 C_PIN object's TryLimit value;
11) If User1 C_PIN Object has a TryLimit Column value >0, then

    Invoke the `StartSession` method with SPID = Locking SP UID, HostSigningAuthority = User1 authority UID, and HostChallenge = a value that does not match the current User1 C_PIN object's PIN column value, until User1 C_PIN object's Tries value = User1 C_PIN object's TryLimit value;

 © TCG 2025

12) Power cycle the SD;

13) If SID C_PIN Object has a TryLimit Column value >0, then

   a. Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;

   b. Invoke the `Get` method on SID Authority's C_PIN Tries Column; and

   c. CLOSE_SESSION;

14) If Admin1 C_PIN Object has a TryLimit Column value >0, then

   a. Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;

   b. Invoke the `Get` method on Admin1 Authority's C_PIN Tries Column; and

   c. CLOSE_SESSION;

   and

15) If User1 C_PIN Object has a TryLimit Column value >0, then

   a. Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;

   b. Invoke the `Get` method on User1 Authority's C_PIN Tries Column; and

   c. CLOSE_SESSION.

Expected Response

1) Step #1, FAIL if the Persistence Column of C_PIN object for any authority (SID/Admin1/User1 to be tested) = TRUE;

2) Steps #2-8 and steps #12-15 SUCCEED;

3) Every `StartSession` method in steps #9, #10, and #11 results in a `SyncSession` method with a status code of NOT_AUTHORIZED;

4) For test step #13, if SID C_PIN TryLimit Column value > 0, then

   a. Admin SP session opens successfully

   b. The `Get` method on SID Authority's C_PIN Tries Column returns 0;

5) For test step #14, if Admin1 C_PIN TryLimit Column value > 0, then

   a. Locking SP session opens successfully

   b. The `Get` method on Admin1 Authority's C_PIN Tries Column returns 0;

   and

6) For test step #15, if User1 C_PIN TryLimit Column value > 0, then

   a. Locking SP session opens successfully

   b. The `Get` method on User1 Authority's C_PIN Tries Column returns 0.

 © TCG 2025

SPF-06: Next

Notes

Start of Informative Comment

Testing of the `Next` method to verify that the method works correctly in multiple conditions.

This test contains two different tests, but only one test is required per SSC, as specified by the SSC applicability section of each test.

End of Informative Comment

Case 1:

SSC Applicability

This test case applies to the following SSCs:

1) Opal 1.00;
2) Opal 2.00;
3) Opal 2.01;
4) Opal 2.02; or
5) Opal 2.30.

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID;
2) Invoke the `Get` method on the `LockingInfo` table's MaxRanges Column;
3) Invoke the `Next` method on the `Locking` table with an empty parameter list;
4) Invoke the `Next` method on the `Locking` table with the Where parameter set to the first UID from the list of UIDs returned in step #3, and the Count parameter set to 1; and
5) CLOSE_SESSION.

Expected Response

1) Steps #1-5 SUCCEED;
2) Step #3
   a. returns a list of UIDs where the number of values = the MaxRanges value + 1, and
   b. the first four bytes of each UID returned are 0x00000802; and
3) Step #4 returns a list that contains only the UID that was second in the list of UIDs returned in Step #3.

 © TCG 2025

Case 2:

SSC Applicability

This test case applies to the following SSCs:

1) Opalite 1.00;
2) Pyrite 1.00;
3) Pyrite 2.00;
4) Pyrite 2.01; or
5) Ruby 1.00.

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID;
2) Invoke the `Next` method on the `MethodID` table with an empty parameter list;
3) Invoke the `Next` method on the `MethodID` table with the Where parameter set to the first UID from the list of UIDs returned in step #3 and the Count parameter set to 1; and
4) CLOSE_SESSION.

Expected Response

1) Steps #1-4 SUCCEED;
2) Step #2
   a. returns a list of UIDs where the number of values >= 7, and
   b. the first four bytes of each UID returned are 0x00000006; and
3) Step #3 returns a list that contains only the second UID from the list of UIDs returned in Step #2.

SPF-07: Host Session Number (HSN)

Notes

Start of Informative Comment

Test the Host Session Number to verify that the SD responses with the corresponding Host Session Number provided by the host.

End of Informative Comment

Prerequisites

None

 © TCG 2025

Test Sequence

1) Invoke the `StartSession` method with HostSessionID = ARBITRARILY_VARYING HSN, SPID = Admin SP UID, and HostSigningAuthority = SID authority UID;
2) Invoke the `Get` method on MSID C_PIN credential's PIN Column; and
3) CLOSE_SESSION.

Expected Response

1) Steps #1-3 SUCCEED;
2) The `StartSession` method in step #1 results in a `SyncSession` method with the same HSN as parameterized in the `StartSession` method; and
3) The Packet received in step #2 that contains the `Get` method response has the same HSN as parameterized in the `StartSession` method.

SPF-08: RevertSP

Notes

Start of Informative Comment

See [2] for support requirements on RevertSP and KeepGlobalRangeKey/KeepData. There are three tests in this test case. Each must be performed.

End of Informative Comment

Case 1:

SSC Applicability

This test applies to all SSCs supported by this specification.

Prerequisites

None

Test Sequence

1) Write the MAGIC_PATTERN over 64 logical blocks beginning at LBA 0;
2) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
3) Invoke the `RevertSP` method with the KeepGlobalRangeKey/KeepData omitted;
4) Invoke the `StartSession` method with SPID = Locking SP UID; and
5) This test step varies based on the SSC version:

 © TCG 2025

a. For all SSCs supported by this specification other than Pyrite 1.00, read 64 logical blocks beginning at LBA 0; and

b. For Pyrite 1.00, do nothing for this step.

Expected Response

1) Steps #1-3 SUCCEED;
2) The `StartSession` method in step #4 results in a `SyncSession` method with a status code of INVALID_PARAMETER; and
3) For all SSCs supported by this specification other than Pyrite 1.00, the Read command in step #5 responds in one of the following ways:
   a. The Read command FAILS and no data is transferred;
   b. The Read command FAILS, data is transferred, and the returned data does not match the MAGIC_PATTERN; or
   c. The Read command SUCCEEDS, data is transferred, and the returned data does not match the MAGIC_PATTERN.

Case 2:

SSC Applicability

This test applies to all SSCs supported by this specification except for Pyrite 1.00.

Prerequisites

None

Test Sequence

1) Write the MAGIC_PATTERN over 64 logical blocks beginning at LBA 0;
2) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
3) Invoke the `RevertSP` method with the KeepGlobalRangeKey/KeepData present and set to FALSE;
4) Invoke the `StartSession` method with SPID = Locking SP UID; and
5) Read 64 logical blocks beginning at LBA 0.

Expected Response

1) Steps #1-3 SUCCEED;
2) The `StartSession` method in step #4 results in a `SyncSession` method with a status code of INVALID_PARAMETER; and
3) The Read command in step #5 responds in one of the following ways:
   a. The Read command FAILS and no data is transferred;

 © TCG 2025

b. The Read command FAILS, data is transferred, and the returned data does not match the MAGIC_PATTERN; or

c. The Read command SUCCEEDS, data is transferred, and the returned data does not match the MAGIC_PATTERN.

Case 3:

SSC Applicability

This test applies to all SSCs supported by this specification except for Pyrite 1.00.

Prerequisites

1) Locking_GlobalRange ReadLockEnabled, WriteLockEnabled, ReadLocked and WriteLocked column values = FALSE; and
2) If non-Global Locking Range objects are implemented, then all non-Global Locking Range objects' ReadLockEnabled, WriteLockEnabled, ReadLocked and WriteLocked column values = FALSE and RangeStart and RangeLength columns = 0.

Test Sequence

1) Write the MAGIC_PATTERN over 64 logical blocks beginning at LBA 0;
2) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
3) Invoke the `RevertSP` method with the KeepGlobalRangeKey/KeepData present and set to TRUE;
4) Invoke the `StartSession` method with SPID = Locking SP UID; and
5) Read 64 logical blocks beginning at LBA 0.

Expected Response

1) Steps #1-3 SUCCEED;
2) The `StartSession` method in step #4 results in a `SyncSession` method with a status code of INVALID_PARAMETER; and
3) The Read command in step #5 returns data that matches the MAGIC_PATTERN.

SPF-09: Range Alignment Verification

Notes

Start of Informative Comment

None

End of Informative Comment

 © TCG 2025

SSC Applicability

This test case only applies to Opal 2.00, 2.01, 2.02 and 2.30, and Ruby 1.00 if the AlignmentRequired column value in the `LockingInfo` table = TRUE.

This test case does not apply to any other SSC.

Prerequisites

1) Confirm the AlignmentRequired column value in the `LockingInfo` table = TRUE. If AlignmentRequired = FALSE do not perform the test and the Test Suite marks the result as NA.

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Invoke the `Get` method on the `LockingInfo` table to retrieve the LogicalBlockSize, AlignmentGranularity and LowestAlignedLBA column values;
3) If AlignmentGranularity is > 1, then invoke the `Set` method on RangeLength and RangeStart columns with RangeStart and RangeLength values satisfying the conditions:
   a. [(RangeStart - LowestAlignedLBA) % AlignmentGranularity] = 0; and
   b. [RangeLength % AlignmentGranularity] = 0;

   and
4) CLOSE_SESSION.

Expected Response

1) If AlignmentGranularity is = 1 then mark the test NA; and
2) If AlignmentGranularity is > 1, steps #1-4 SUCCEED

SPF-10: Byte Table Access Granularity

Notes

Start of Informative Comment

None

End of Informative Comment

SSC Applicability

This test case only applies to the following SSCs:

1) Opal 2.00;
2) Opal 2.01;
3) Opal 2.02;

 © TCG 2025

4) Opal 2.30;
5) Opalite 1.00;
6) Pyrite 1.00;
7) Pyrite 2.00;
8) Pyrite 2.01; or
9) Ruby 1.00.

Prerequisites

1) Confirm the MandatoryWriteGranularity column value of the `DataStore` table > 1. If the MandatoryWriteGranularity column value = 1, do not perform the test and the Test Suite marks the result as NA.

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Invoke the `Get` method on the DataStore object in the `Table` table to retrieve the MandatoryWriteGranularity column value;
3) Invoke the `Set` method to write the `DataStore` table with a number of 0s = a non-zero multiple of the MandatoryWriteGranularity column value; and
4) CLOSE_SESSION.

Expected Response

1) Steps #1-4 SUCCEED

SPF-11: Stack Reset

Notes

Start of Informative Comment

Reference SD vendor documentation to determine whether the command is supported.

End of Informative Comment

Prerequisites

1) User1 is not enabled

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Send a subpacket that contains a StartTransaction token with a status code of 0x00;

 © TCG 2025

3) Invoke the `Set` method on the Enabled Column of User1 Authority with a value of TRUE;
4) Issue STACK_RESET command;
5) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
6) Invoke the `Get` method to retrieve the value of the Enabled Column of User1 Authority; and
7) CLOSE_SESSION.

Expected Response

1) Steps #1-7 SUCCEED; and
2) The `Get` method in step #6 returns a value of FALSE.

SPF-12: TPer Reset

Notes

Start of Informative Comment

None

End of Informative Comment

Case 1:

SSC Applicability

This test case only applies to the following SSCs:

1) Opal 2.00;
2) Opal 2.01;
3) Opal 2.02;
4) Opal 2.30;
5) Opalite 1.00;
6) Pyrite 1.00;
7) Pyrite 2.00;
8) Pyrite 2.01; or
9) Ruby 1.00.

Prerequisites

1) ProgrammaticResetEnable set to TRUE;
2) Locking_GlobalRange has ReadLocked and WriteLocked columns set to FALSE;
3) Locking_GlobalRange has ReadLockEnabled and WriteLockEnabled columns are set to TRUE;
4) LockOnReset column value includes Programmatic;

 © TCG 2025

5) For Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00, if the MBR Shadowing feature is supported, then the Enable column value of the `MBRControl` table = FALSE; and
6) For Opal 1.00, 2.00, 2.01, 2.02 and 2.30, and Opalite 1.00, the Enable column value of the `MBRControl` table = FALSE.

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Issue the TPER_ RESET command;
3) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
4) Invoke the `Get` method on the Locking_GlobalRange ReadLocked and WriteLocked columns;
5) CLOSE_SESSION;
6) Write the MAGIC_PATTERN over an ARBITRARILY_VARYING_LBA_RANGE; and
7) Read from the same ARBITRARILY_VARYING_LBA_RANGE as in Step #6.

Expected Response

1) Steps #1-5 SUCCEED;
2) The `Get` method in step #4 returns values of TRUE;
3) The Write command in Step #5 returns an interface status of Data Protection Error; and
4) The Read command in Step #6 returns an interface status of Data Protection Error.

Case 2:

SSC Applicability

This test case applies to the following SSCs:

1) Opal 2.00;
2) Opal 2.01;
3) Opal 2.02;
4) Opal 2.30; or
5) Opalite 1.00.

This test case applies to all other SSCs supported by this specification only if the MBR Shadowing feature is supported.

Prerequisites

1) ProgrammaticResetEnable column is set to TRUE;
2) Locking_GlobalRange has ReadLockEnabled and WriteLockEnabled columns are set to FALSE;

 © TCG 2025

3) DoneOnReset column value includes Programmatic;
4) Done column is set to TRUE;
5) First 64 logical blocks of `MBR` table have been set with MAGIC_PATTERN; and
6) If the MBR Shadowing feature is supported, the Enable column value of the `MBRControl` table = TRUE.

Test Sequence

1) Write 0s to 64 logical blocks beginning at LBA 0;
2) Issue the TPER_RESET command;
3) Write 1s to 64 logical blocks beginning at LBA 0;
4) Read 64 logical blocks beginning at LBA 0;
5) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID; and
6) Invoke the `Get` method on the MBRControl Done column.

Expected Response

1) Steps #1-2 SUCCEED;
2) The Write command in Step #3 returns an interface status of Data Protection Error;
3) The Read command in Step #4 returns data that matches the MAGIC_PATTERN;
4) Steps #5-6 SUCCEED; and
5) The `Get` method in step #6 returns the value FALSE.

SPF-13: Authenticate

Notes

Start of Informative Comment

None

End of Informative Comment

SSC Applicability

This test case only applies to the following SSCs:

1) Opal 2.00;
2) Opal 2.01;
3) Opal 2.02;
4) Opal 2.30;
5) Opalite 1.00;
6) Pyrite 1.00;

 © TCG 2025

7) Pyrite 2.00;
8) Pyrite 2.01; or
9) Ruby 1.00.

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Admin SP UID;
2) Invoke the `Authenticate` method with Authority = SID Authority UID and Proof = C_PIN_SID PIN column value;
3) Invoke the `Get` method on UID Column of SID C_PIN; and
4) CLOSE_SESSION.

Expected Response

1) Steps #1-4 SUCCEED; and
2) The `Get` method in step #3 returns the C_PIN_SID PIN object's UID column value.

SPF-14: Session Abort (Deprecated)

This test case has been removed due to similar functionality being tested elsewhere. This section will be removed in a future version of this specification.

SPF-15: Random

Notes

Start of Informative Comment

The quality of RNG is not in the scope of this test.

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID;
2) Invoke the `Random` method with a Count = 32;
3) Invoke the `Random` method with a Count = 32; and

 © TCG 2025

4) CLOSE_SESSION.

Expected Response

1) Steps #1-4 SUCCEED;
2) The value returned by the `Random` method in step #2 is 32 bytes long and does not contain either all 0s or all 1s;
3) The value returned from the `Random` method in step #3 is 32 bytes long and does not contain either all 0s or all 1s; and
4) The two values returned from the `Random` method in steps #2 and #3 are different.

SPF-16: CommonName

Notes

Start of Informative Comment

None

End of Informative Comment

SSC Applicability

This test case only applies to the following SSCs:

1) Opal 2.00;
2) Opal 2.01;
3) Opal 2.02;
4) Opal 2.30; or
5) Ruby 1.00.

Prerequisites

1) Admin1 is enabled

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Invoke the `Set` method on the CommonName column of the Admin1 authority object using the MAGIC_PATTERN;
3) Invoke the `Set` method on the CommonName column of Locking_GlobalRange using the MAGIC_PATTERN;
4) Invoke the `Get` method on the CommonName column of the Admin1 authority object;
5) Invoke the `Get` method on the CommonName column of Locking_GlobalRange; and
6) CLOSE_SESSION.

 © TCG 2025

Expected Response

1) Steps #1-6 SUCCEED; and
2) The values returned from the `Get` methods in steps #4-5 are the same as the values previously `Set` in steps #2-3.

SPF-17: Additional DataStore Tables

Notes

Start of Informative Comment

Only one of the following tests is performed based on the value of the Maximum Number of DataStore tables field in the DataStore table Feature Descriptor.

End of Informative Comment

Case 1:

SSC Applicability

This test case applies to the following SSCs:

1) Opal 2.00;
2) Opal 2.01;
3) Opal 2.02;
4) Opal 2.30; or
5) All other SSCs supported by this specification, if the Additional DataStore Tables Feature Set is implemented.

Prerequisites

1) In the DataStore table Feature Descriptor, the Maximum Number of DataStore Tables field value = 1; and
2) Locking SP is in the Manufacture-Inactive State.

Test Sequence

1) Issue Level 0 Discovery command to retrieve the DataStore Table Size Alignment field;
2) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
3) Invoke the `Activate` method on the Locking SP with a DataStoreTableSize parameter value = the value of the DataStore Table Size Alignment field of the Level 0 Discovery Feature Descriptor;
4) CLOSE_SESSION;
5) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
6) Invoke the `Get` method to retrieve the `DataStore` table's Rows column value from the Table table; and

 © TCG 2025

7) CLOSE_SESSION.

Expected Response

1) Steps #1-7 SUCCEED; and
2) The `Get` method in step #6 returns a value equal to the DataStoreTableSize parameter value in step #3.

Case 2:

SSC Applicability

This test case applies to the following SSCs:

1) Opal 2.00;
2) Opal 2.01;
3) Opal 2.02;
4) Opal 2.30; or
5) All other SSCs supported by this specification, if the Additional DataStore Tables Feature Set is implemented.

Prerequisites

1) In the DataStore Table Feature Descriptor, the Maximum Number of DataStore tables field value > 1; and
2) Locking SP is in the Manufactured-Inactive State.

Test Sequence

1) Issue Level 0 Discovery command to retrieve the DataStore Table Size Alignment field;
2) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
3) Invoke the `Activate` method with a DataStoreTableSize parameter value containing a number of items = the Maximum Number of DataStore Tables field, with values = the value of the DataStore Table Size Alignment field of the Level 0 Discovery Feature Descriptor;
4) CLOSE_SESSION;
5) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
6) Invoke the `Get` method to retrieve each `DataStore` table's Rows column value from the `Table` table; and
7) CLOSE_SESSION.

Expected Response

1) Steps #1-5 SUCCEED; and
2) For each `DataStore` table, the `Get` method in step #6 returns a value equal to the DataStoreTableSize parameter value in step #3.

 © TCG 2025

SPF-18: Range Crossing Behavior

Notes

Start of Informative Comment

Test that the range crossing behavior is as specified by the returned value for range crossing.

Determine support for feature via Level 0 Discovery.

End of Informative Comment

SSC Applicability

This test case applies to the following SSCs:

1) Opal 1.00;
2) Opal 2.00;
3) Opal 2.01;
4) Opal 2.02;
5) Opal 2.30; or
6) All other SSCs supported by this specification, if Locking_Range1 is implemented

Prerequisites

1) Locking_Range1 length is non-zero and does not span the entire SD; and
2) Locking_GlobalRange and Locking_Range1 are unlocked.

Test Sequence

1) Issue a Write command with the MAGIC_PATTERN, with a beginning LBA in Locking_Range1 and ending LBA in Locking_GlobalRange. For ZNS device, if Zone Capacity is less than Zone Size, skip this step; and
2) Issue a Read command, with a beginning LBA in Locking_Range1 and ending LBA in Locking_GlobalRange. For ZNS device, if Read Across Zone Boundaries is cleared to zero, skip this step.

Expected Response

1) If Range Crossing is supported and if step #1 or #2 is not skipped, then steps #1-2 SUCCEED; and
2) If Range Crossing is not supported and if step #1 or #2 is not skipped, then steps #1-2 FAIL. The Write command in step #1 and the Read command in step #2 return an interface status of Other Invalid Command Parameter.

SPF-19: Block SID Authentication

Notes

Start of Informative Comment

None

 © TCG 2025

End of Informative Comment

SSC Applicability

This test case applies to the following SSCs:

1) Opalite 1.00;
2) Pyrite 1.00;
3) Pyrite 2.00;
4) Pyrite 2.01;
5) Ruby 1.00;
6) Opal 2.02;
7) Opal 2.30; or
8) All other SSCs supported by this specification, if the Block SID Feature Set is implemented.

Prerequisites

1) SID C_PIN credential is the same as the value of the MSID C_PIN credential.

Test Sequence

1) Issue IF-SEND with the following parameters:
   a. Protocol ID = 0x02;
   b. ComID = 0x0005; and
   c. Hardware Reset bit in Clear Events field = 1;
2) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
3) Issue an IF-RECV Level 0 Discovery with the following conditions:
   a. Security Protocol = 1;
   b. Security Protocol Specific = 0x0001; and
   c. Transfer Length is a value large enough to retrieve the entire response data of Level 0 Discovery;
4) Trigger a TCG Storage Hardware Reset on the SD;
5) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
6) Issue IF-SEND with the following parameters:
   a. Protocol ID = 0x02;
   b. ComID = 0x0005; and
   c. Hardware Reset = 0;
7) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
8) Issue an IF-RECV Level 0 Discovery with the following conditions:
   a. Security Protocol = 1;

 © TCG 2025

b. Security Protocol Specific = 0x0001; and

c. Transfer Length is a value large enough to retrieve the entire response data of Level 0 Discovery;

9) Power cycle the SD; and

10) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID.

Expected Response

1) Steps #1 and #3-10 SUCCEED;

2) The `StartSession` method in step #2 results in a `SyncSession` method with a status code of NOT_AUTHORIZED;

3) The Hardware Reset field in the Block SID Authentication Feature Descriptor in the Level 0 Discovery response returned in #3 = 1;

4) The `StartSession` method in step #7 results in a `SyncSession` method with a status code of NOT_AUTHORIZED; and

5) The Hardware Reset field in the Block SID Authentication Feature Descriptor in the Level 0 Discovery response returned in #8 = 0.

SPF-20: Data Removal Mechanism

Start of Informative Comment

Test the `Set` method and the `Get` method on the ActiveDataRemovalMechanism column in the Data Removal Mechanism table to make sure this table is functional

End of Informative Comment

SSC Applicability

This test case applies to the following SSCs:

1) Pyrite 2.00;

2) Pyrite 2.01; or

3) All other SSCs supported by this specification, if the Data Removal Mechanism feature is implemented.

Prerequisites

1) If DUT is other than Opal v2.02 or later versions, knowledge of supported Data Removal Mechanisms from Supported Data Removal Mechanisms Feature Descriptor in Level 0 Discovery

Test Sequence

1) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;

2) Invoke the `Get` method on the ActiveDataRemovalMechanism column of the `DataRemovalMechanism` table;

3) Invoke the `Set` method on the ActiveDataRemovalMechanism column of the `DataRemovalMechanism` table,

a. For Opal 2.02 or later versions, with Cryptographic Erase Data Removal Mechanism; and

 © TCG 2025

b. For other SSCs, with one of the supported Data Removal Mechanisms returned in Level 0 Discovery;

4) CLOSE_SESSION;
5) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = Anybody authority UID;
6) Invoke the `Get` method on the ActiveDataRemovalMechanism column of the `DataRemovalMechanism` table; and
7) CLOSE_SESSION.

Expected Response

1) Steps #1-7 SUCCEED;
2) The value returned from the `Get` method in Step #2 matches is equal to one of the bits set in the Supported Data Removal Mechanisms returned in Level 0 Discovery; and
3) The value returned from the `Get` method in Step #6 matches the value that was set in Step #3.

 © TCG 2025

7 Error Test Cases

7.1 Introduction

The goal of this section is twofold: a) to reduce the overall number of error tests, and b) to require only a single instance of a common error test in the test cases. All possible unique error responses defined in the SSC specifications are included in at least one test case.

Unless otherwise noted within a specific test case, session status is deemed to remain unaffected by the performance of any tests in this section.

This section does not include any tests where multiple errors are encoded in a payload from the host. Each test case only tests for a single error condition; however, some test cases can result in different possible error responses.

For every test case in this specification that specifies an error status code response, session abort is an acceptable response. In the case of session abort, the SD sending a CloseSession response is acceptable.

7.2 Common Prerequisites

Unless otherwise noted, the following set of prerequisites apply for each test in this section:

1. SD is in Awaiting IF-SEND;
2. Locking SP is in Manufactured state;
3. The values of any credentials used are known;
4. All sessions are Read-Write sessions; and
5. No sessions are open.

ETC-01: Native Protocol Read/Write Locked Error Responses

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

1) Locking_GlobalRange ReadLockEnabled, WriteLockEnabled, ReadLocked and WriteLocked column values = TRUE;
2) If non-Global Locking Range objects are implemented, then all non-Global Locking Range objects ReadLockEnabled, WriteLockEnabled, ReadLocked and WriteLocked column values = FALSE and RangeStart and RangeLength columns values = 0;
3) For Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00, if the MBR Shadowing feature is supported, then the Enable column value of the `MBRControl` table = FALSE; and
4) For Opal 1.00, 2.00, 2.01, 2.02 and 2.30, and Opalite 1.00, the Enable column value of the `MBRControl` table = FALSE.

 © TCG 2025

Test Sequence

1) Issue each of the Write commands (as identified by [2]) that are supported by the SD and the Test Suite. If an LBA range is required for a supported command, write to an ARBITRARILY_VARYING_LBA_RANGE. If other parameters are required for a supported command, use ARBITRARILY_VARYING_COMMAND_PARAMETERS. Refer to section 4.6; and
2) Issue each of the Read commands (as identified by [2]) that are supported by the SD and the Test Suite. If an LBA range is required for a supported command, read from an ARBITRARILY_VARYING_LBA_RANGE. If other parameters are required for a supported command, use ARBITRARILY_VARYING_COMMAND_PARAMETERS. Refer to section 4.6.

Expected Response

1) Each of the issued commands in Steps #1-2 FAIL; and
2) For all supported Write commands in step #1 and all supported Read commands in step #2, the SD responds with:
   a. no data transfer; and
   b. return an interface status of Data Protection Error, (See [2]).

ETC-02: General – IF-SEND/IF-RECV Synchronous Protocol

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `Properties` method within an IF-SEND using a valid ComID and do not retrieve the response with an IF-RECV; and
2) Invoke the `Properties` method using the ComID from the previous step.

Expected Response

1) Step #1 SUCCEEDS; and
2) Step #2 FAILS. The IF-SEND command returns an interface status of Synchronous Protocol Violation.

 © TCG 2025

ETC-03: Invalid IF-SEND Transfer Length

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `Properties` method to determine SD's MaxComPacketSize; and
2) Invoke the `Properties` method with the correct ComPacket Header Length field to match the required ComPacket payload size but with the IF-SEND Transfer Length set to a value > MaxComPacketSize.

Expected Responses

1) Step #1 SUCCEEDS; and
2) The IF-SEND in step #2 fails with a result of "Invalid Transfer Length parameter on IF-SEND".

ETC-04: Invalid SessionID - Regular Session

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Admin SP UID;
2) Invoke the `Get` method on MSID's credential object in `C_PIN` table with a Packet SessionID value <> the current SessionID value; and
3) CLOSE_SESSION.

 © TCG 2025

Expected Responses

1) Steps #1-3 SUCCEED; and
2) IF-RECV in step #2 has a ComPacket header value of “All Response(s) returned - no further data”, (See [1]).

ETC-05: Unexpected Token Outside of Method – Regular Session

Notes

Start of Informative Comment

This test verifies the condition corresponding to [1], Section 3.2.2.4.2 item 2. The reason for the expected response #2 of “All Response(s) returned - no further data” is that the device is in the “Awaiting IF_SEND” state, see [1], Section 3.3.10.5

End of Informative Comment

Prerequisites

1) User1 authority object’s Enabled Column is set to TRUE

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Invoke the `Set` method on the Enabled Column of User1 Authority with a value of FALSE and EndList Token before the Call Token;
3) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
4) Invoke the `Get` method on the Enabled Column of User1 Authority; and
5) CLOSE_SESSION.

Expected Response

1) Step #1 SUCCEEDS;
2) IF-RECV in step #2 has a ComPacket header value of “All Response(s) returned - no further data” (See [1]), or returns a ComPacket with a `CloseSession` method;
3) Step #3-5 SUCCEED; and
4) Step #4 the `Get` method on the Enabled Column of the User1 Authority returns a value of TRUE.

ETC-06: Unexpected Token in Method Header – Regular Session

Notes

Start of Informative Comment

This test verifies the condition corresponding to [1], Section 3.2.2.4.2 item 3.

 © TCG 2025

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Invoke the `Set` method on the Enabled Column of User1 Authority with a value of FALSE and an EndList Token immediately after the Call Token; and
3) CLOSE_SESSION.

Expected Response

1) Step #1 SUCCEEDS;
2) Step #2 `Set` method returns a status code of NOT_AUTHORIZED, or returns a ComPacket with a `CloseSession` method; and
3) Step #3 SUCCEEDS if step #2 returns a status code of NOT_AUTHORIZED.

ETC-07: Unexpected Token Outside of Method – Control Session

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and an EndList Token before the Call Token; and
2) Invoke the `StartSession` method with SPID = Locking SP UID.

Expected Response

1) IF-RECV in step #1 has a ComPacket header value of “All Response(s) returned - no further data”, (See [1]); and
2) Steps #2 SUCCEEDS.

 © TCG 2025

ETC-08: Unexpected Token in the Method Parameter List – Control Session

Notes

Start of Informative Comment

This test verifies the condition corresponding to [1], Section 3.2.2.4.2 items 1 and 4. The reason for the expected response #1a of "All Response(s) returned - no further data" is the description in [1], Please refer to the Core Specification Section 3.3.7.1.5.

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `Properties` method with StartList immediately after the Parameter StartList

Expected Response

1) One of the following responses is generated:
   a. IF-RECV in step #1 has a ComPacket header value of "All Response(s) returned - no further data", (See [1]); or
   b. The `Properties` method in step #1 returns a status code of INVALID_PARAMETER.

ETC-09: Exceeding Transaction Limit

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `Properties` method to identify MaxTransactionLimit;
2) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
3) Send a subpacket that contains MaxTransactionLimit + 1 StartTransaction Tokens;

 © TCG 2025

4) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID; and
5) CLOSE_SESSION.

Expected Response

1) Steps #1-2 SUCCEED;
2) IF-RECV in step #3 has a ComPacket header value of “All Response(s) returned - no further data” (See [1]), or returns a ComPacket with a `CloseSession` method; and
3) Steps #4-5 SUCCEED.

ETC-10: Invalid Invoking ID - Get

Notes

Start of Informative Comment

The `LockingInfo` table is a single row table. The UID used in the following test refers to row 5, a nonexistent row of the `LockingInfo` table.

This test case tests the following requirement from [1]:

“Unless otherwise noted in a method's description, this status code (NOT_AUTHORIZED) SHALL be returned whenever there is no row in the `AccessControl` table to represent the InvokingID/MethodID combination, or when there is a row but the ACL for the InvokingID/MethodID combination has not been satisfied.”

End of Informative Comment

Case 1:

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Invoke the `Get` method on Invoking UID of 00 00 08 01 AA BB CC DD; and
3) CLOSE_SESSION.

Expected Response

1) Step #1 SUCCEEDS;
2) The `Get` method in step #2 returns a status code of NOT_AUTHORIZED; and
3) Step #3 SUCCEEDS.

 © TCG 2025

Case 2:

Notes

Start of Informative Comment

This test validates correct behavior when the `Get` method is invoked on a Byte table and the authority does not have access to retrieve contents from the byte table.

This test case tests the following requirement from [1]:

"If the currently authenticated authorities do not satisfy the access control restrictions for invoking the `Get` method on a byte table, the method SHALL return an empty results list."

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Anybody authority UID;
2) Invoke the `Get` method on Invoking UID of 00 00 10 01 00 00 00 00 (`DataStore` table); and
3) CLOSE_SESSION.

Expected Response

1) Steps #1 SUCCEEDS;
2) The `Get` method in step #2 returns a status code of NOT_AUTHORIZED or SUCCESS and an empty results list; and
3) Step #3 SUCCEEDS.

Case 3:

Notes

Start of Informative Comment

This test validates correct behavior when the `Get` method is invoked on an Object table and the authority does not have access to retrieve contents from the Object table.

This test case tests the following requirement from [1]:

"When the `Get` method is invoked on a table or object, only the values that are readable based on currently authenticated authorities and their associated ACE restrictions for the method SHALL be returned. "

Cell values that have been requested but are not permitted to be read by the currently authenticated authorities are not returned. Since the return value of the method for non-byte tables is a list of name-value pairs, cells, which the host invoking the `Get` method does not have access to, are omitted from the return result. [If a column is known to exist but not returned with a value, then the host is able to discern that it did not have permission to invoke the `Get` method on that cell. It is not an error to request columns that are not permitted to be retrieved.

End of Informative Comment

 © TCG 2025

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Invoke the `Get` method on the InvokingID 00 00 00 0B 00 01 00 01 (C_PIN_Admin1) to get the PIN, CharSet, TryLimit, and Tries columns; and
3) CLOSE_SESSION.

Expected Response

1) Steps #1 SUCCEEDS;
2) The `Get` method in step #2 returns a status code of SUCCESS and only returns the CharSet, TryLimit, and Tries column values; and
3) Step #3 SUCCEEDS.

Case 4:

Notes

Start of Informative Comment

This test validates correct behavior when the `Get` method is invoked on a non-table UID.

This test case is similar to Test Case 1, but instead it tests with a valid InvokingUID but there is no row in the `ACL` table that matches the InvokingID/MethodID combination.

This test case tests the following requirement from [1]:

“Unless otherwise noted in a method's description, this status code (NOT_AUTHORIZED) SHALL be returned whenever there is no row in the `AccessControl` table to represent the InvokingID/MethodID combination, or when there is a row but the ACL for the InvokingID/MethodID combination has not been satisfied.”

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Anybody authority UID;
2) Invoke the `Get` method on the InvokingID 00 00 00 00 00 00 00 01 (ThisSP); and
3) CLOSE_SESSION.

 © TCG 2025

Expected Response

1) Steps #1 SUCCEEDS;
2) The `Get` method in step #2 returns a status code of NOT_AUTHORIZED and an empty results list; and
3) Step #3 SUCCEEDS.

ETC-11: Invalid Invoking ID – Non-Get

Notes

Start of Informative Comment

The `LockingInfo` table is a single row table. The UID used in the following test refers to row 5, a non-existing row of the `LockingInfo` table.

This test uses `Set` method to represent all non-`Get` methods.

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID;
2) Invoke the `Set` method on Invoking UID of 00 00 08 01 00 00 00 05; and
3) CLOSE_SESSION.

Expected Response

1) Steps #1 SUCCEEDS;
2) The `Set` method in step #2 returns a status code of NOT_AUTHORIZED; and
3) Step #3 SUCCEEDS.

ETC-12: Authorization

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

None

 © TCG 2025

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID;
2) Invoke the `Set` method on the Enabled column of the User1 Authority; and
3) CLOSE_SESSION.

Expected Response

1) Steps #1 SUCCEEDS;
2) The `Set` method in step #2 returns a status code of NOT_AUTHORIZED; and
3) Step #3 SUCCEEDS.

ETC-13: Malformed ComPacket Header – Regular Session

Notes

Start of Informative Comment

This tests a malformed Length field in the ComPacket header whereas TRANSFER LENGTH field in IF-SEND CDB has a correct value. If it is not possible to invoke a `Set` method that exceeds the TPer's MaxComPacketSize, then this test cannot be performed and the result should be marked as NA.

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `Properties` method to identify the MaxComPacketSize;
2) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
3) This test step varies based on SSC version:
   a. For Opal 1.00, invoke the `Set` method on the `MBR` table, such that the Length field in the ComPacket header exceeds the TPer's MaxComPacketSize – 20 (where 20 is the length of ComPacket header field), and the IF-SEND Transfer Length set to a value <= MaxComPacketSize; and
   b. For all SSCs supported by this specification other than Opal 1.00, invoke the `Set` method on the `DataStore` table such that the Length field in the ComPacket header exceeds the TPer's MaxComPacketSize – 20 (where 20 is the length of ComPacket header field), and the IF-SEND Transfer Length set to a value <= MaxComPacketSize;

   and
4) Issue IF-RECV.

 © TCG 2025

Expected Response

1) Steps #1-2 SUCCEED;
2) The IF_SEND in step #3:
   a. SUCCEEDS; or
   b. FAILS with a result of “Invalid Transfer Length parameter on IF-SEND”;
3) [1]The IF-RECV in step #4 returns a ComPacket header with a value of “All Response(s) returned - no further data” (See [1]), or returns a ComPacket with a `CloseSession` method.

ETC-14: Exceed TPer Properties – Regular Session

Notes

Start of Informative Comment

Tests for MaxSubPackets exceeded.

This test verifies the condition corresponding to [1], Section 5.2.2.4.1.1. The reason for the expected response #1a of “All Response(s) returned - no further data” is the description in [1], Section 3.3.7.1.5: “The Host or TPer is free at any time to end a session in which it is participating, but only the host SHALL end the session successfully.”

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `Properties` method to identify the MaxSubPackets;
2) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
3) Send a packet with MaxSubPackets +1 SubPackets. Each SubPacket contains an invocation of the `Set` method on the `DataStore` table;
4) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID; and
5) CLOSE_SESSION.

Expected Response

1) Steps #1-2 SUCCEED;
2) IF-RECV in step #3 has a ComPacket header value of “All Response(s) returned - no further data” (See [1]), or returns a ComPacket with a `CloseSession` method; and
3) Steps #4-5 SUCCEED.

 © TCG 2025

ETC-15: Exceed TPer Properties – Control Session

Notes

Start of Informative Comment

Tests for MaxSubPackets exceeded.

End of Informative Comment

Prerequisites

None

Test Sequence

1) Invoke the `Properties` method to identify the MaxSubPackets; and
2) Invoke the `Properties` method with MaxSubPackets +1 SubPackets. Each SubPacket contains an invocation of the `Properties` Method.

Expected Response

1) Step #1 SUCCEEDS; and
2) IF-RECV in step #2 has a ComPacket header value of “All Response(s) returned - no further data”, (See [1])

ETC-16: Overlapping Locking Ranges

Notes

Start of Informative Comment

None

End of Informative Comment

SSC Applicability

This test case applies to the following SSCs:

1) Opal 1.00, 2.00 and 2.01; or
2) All other SSCs supported by this specification, if the MaxRanges column value in the `LockingInfo` table is > 1.

Prerequisites

None

 © TCG 2025

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Invoke the `Set` method on Locking_Range1. Configure the locking range as follows:
   a. RangeStart = 0;
   b. RangeLength = 64; and
   c. If any SSC supported by this specification other than Opal 1.00, adjust RangeStart and RangeLength according to the Range Start Alignment and Range Length Alignment;
3) Invoke the `Set` method on Locking_Range2. Configure the locking range as follows:
   a. RangeStart = 0;
   b. RangeLength = 64; and
   c. If any SSC supported by this specification other than Opal 1.00, adjust RangeStart and RangeLength according to the Range Start Alignment and Range Length Alignment;

   and
4) CLOSE_SESSION.

Expected Response

1) Steps #1-2 SUCCEED;
2) The `Set` method in step #3 returns a status code of INVALID_PARAMETER; and
3) Step #4 SUCCEEDS.

ETC-17: Invalid Type

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

None

Test Sequences

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Invoke the `Set` method on the Enabled column of the User1 Authority to value of 0xAAAA; and
3) CLOSE_SESSION.

 © TCG 2025

Expected Response

1) Steps #1 SUCCEEDS;
2) The `Set` method in step #2 returns a status code of INVALID_PARAMETER; and
3) Step #3 SUCCEEDS.

ETC-18: RevertSP – GlobalRange Locked

Notes

Start of Informative Comment

None

End of Informative Comment

SSC Applicability

This test case applies to all SSCs supported by this specification other than Pyrite 1.00

Prerequisites

None

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Invoke the `Set` method on GlobalRange with the following conditions:
   a. ReadLockedEnabled = TRUE;
   b. WriteLockedEnabled = TRUE;
   c. ReadLocked = TRUE; and
   d. WriteLocked = TRUE;
3) Invoke the `RevertSP` method on the Locking SP with KeepGlobalRangeKey/KeepData = TRUE; and
4) CLOSE_SESSION.

Expected Response

1) Steps #1-2 SUCCEED;
2) Step #3 `RevertSP` method returns a status code of FAIL; and
3) Step #4 SUCCEEDS.

 © TCG 2025

ETC-19: Activate / ATA Security Interaction

Notes

Start of Informative Comment

See[2]

End of Informative Comment

Prerequisites

1) SID's PIN column value is set to <SID_PASSWORD> value in the SID's C_PIN credential PIN column
2) The ATA Security Feature Set Security Characteristics are Enabled and Unlocked
3) Locking SP is in the Manufactured-Inactive state

Test Sequence

1) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
2) Invoke the `Activate` method on Locking SP object; and
3) CLOSE_SESSION.

Expected Response

1) Step #1 SUCCEEDS;
2) Step #2 the `Activate` method returns a status code of FAIL; and
3) Step #3 SUCCEEDS.

ETC-20: StartSession on Inactive Locking SP

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

1) Locking SP is in the Manufactured-Inactive state

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID.

 © TCG 2025

Expected Response

1) The `StartSession` method in step #1 results in a `SyncSession` method with a status code of INVALID_PARAMETER.

ETC-21: StartSession with Incorrect HostChallenge

Notes

Start of Informative Comment

None

End of Informative Comment

Prerequisites

1) The C_PIN credential associated with Admin1 has a TryLimit column value of 0; or a Tries column value < the TryLimit column value

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID, HostSigningAuthority = Admin1 authority UID, and HostChallenge = a value that is different from the C_PIN_Admin1 PIN column value.

Expected Response

1) The `StartSession` method in step #1 results in a `SyncSession` method with a status code of NOT_AUTHORIZED.

ETC-22: Multiple Sessions

Notes

Start of Informative Comment

There are two tests performed regarding multiple sessions:

Case 1 attempts to start two Read-Write sessions with the Locking SP

Case 2 attempts to start MaxSessions + 1 Read-Only sessions with the Locking SP

End of Informative Comment

Case 1:

Prerequisites

1) No open sessions exist at the start of the Test Sequence

Test Sequence

1) Invoke the `Properties` method to identify the MaxSessions;

 © TCG 2025

2) Invoke the `StartSession` method with SPID = Locking SP UID and Write = TRUE; and
3) Invoke the `StartSession` method with SPID = Locking SP UID and Write = TRUE.

Expected Response

1) Step #1-2 SUCCEEDS; and
2) The `StartSession` method in step #3 results in a `SyncSession` method with a status code of:
   a. If MaxSessions = 1: SP_BUSY or NO_SESSIONS_AVAILABLE
   b. If MaxSession <> 1: SP_BUSY.

Case 2:

Prerequisites

1) No open sessions exist at the start of the Test Sequence

Test Sequence

1) Invoke the `Properties` method to identify the MaxSessions and MaxReadSessions. If MaxSessions = 0 or MaxReadSessions = 0 or MaxReadSessions is omitted, do not perform this test and the Test Suite marks the result as NA;
2) Invoke the `StartSession` method with SPID = Locking SP UID and Write = FALSE up to the lesser of MaxSessions or MaxReadSessions; and
3) Invoke the `StartSession` method with SPID = Locking SP UID and Write = FALSE.

Expected Response

1) Step #1 SUCCEEDS;
2) Every `StartSession` method invoked in step #2 results in a `SyncSession` method with a status code of SUCCESS; and
3) The `StartSession` method in step #3 results in a `SyncSession` method with a status code of NO_SESSIONS_AVAILABLE.

ETC-23: Data Removal Mechanism – Set Unsupported Value

Notes

Start of Informative Comment

Test the `Set` method on the ActiveDataRemovalMechanism column in the Data Removal Mechanism table with an invalid value to make sure a proper error is returned

End of Informative Comment

 © TCG 2025

SSC Applicability

This test case applies to the following SSCs:

1) Pyrite 2.00;
2) Pyrite 2.01; or
3) All other SSCs supported by this specification, if the Data Removal Mechanism feature is implemented.

Prerequisites

1) Knowledge of supported Data Removal Mechanisms from Supported Data Removal Mechanisms Feature Descriptor in Level 0 Discovery

Test Sequence

1) Invoke the `StartSession` method with SPID = Admin SP UID and HostSigningAuthority = SID authority UID;
2) Invoke the `Set` method on the ActiveDataRemovalMechanism column of the DataRemovalMechanism table, with a value that is not one of the supported Data Removal Mechanisms returned in Level 0 Discovery; and
3) CLOSE_SESSION.

Expected Response

1) Steps #1-3 SUCCEED; and
2) The `Set` method in Step #2 returns a status code of INVALID_PARAMETER.

ETC-24: Read Locked and Write Locked Error Responses

Notes

Start of Informative Comment

None

End of Informative Comment

SSC Applicability

This test case applies to the following SSCs:

1) Opal 1.00;
2) Opal 2.00;
3) Opal 2.01;
4) Opal 2.02;
5) Opal 2.30; or
6) All other SSCs supported by this specification, if Locking_Range1 is implemented.

 © TCG 2025

Prerequisites

1) Locking_GlobalRange ReadLockEnabled, WriteLockEnabled, ReadLocked and WriteLocked column values = TRUE;
2) If non-Global Locking Range objects are implemented, then all non-Global Locking Range objects ReadLockEnabled, WriteLockEnabled, ReadLocked and WriteLocked column values = FALSE and RangeStart and RangeLength columns values = 0;
3) For Pyrite 1.00, 2.00 and 2.01, and Ruby 1.00, if the MBR Shadowing feature is supported, then the Enable column value of the `MBRControl` table = FALSE; and
4) For Opal 1.00, 2.00, 2.01, 2.02 and 2.30, and Opalite 1.00, the Enable column value of the `MBRControl` table = FALSE.

The Read command and the Write command (as identified by [2]) issued in this test sequence are the commands that are supported by the SD and by the Test Suite. The LBA range of the Read command and the Write command is defined in the ARBITRARILY_VARYING_LBA_RANGE. If other parameters are required for a supported command, use ARBITRARILY_VARYING_COMMAND_PARAMETERS. Refer to section 4.6.

Test Sequence

1) Invoke the `StartSession` method with SPID = Locking SP UID and HostSigningAuthority = Admin1 authority UID;
2) Invoke the `Set` method on Locking_Range1. Configure the locking range as follows:
   a. RangeStart = 0;
   b. RangeLength = 1024; and
   c. If the SD supports an SSC other than Opal 1.00, adjust RangeStart and RangeLength according to the Range Start Alignment and Range Length Alignment;
3) Invoke the `Set` method to configure Locking_Range1 ReadLockEnabled and WriteLockEnabled column values = TRUE, ReadLocked column value = FALSE and WriteLocked column values = TRUE;
4) Issue a Read command in Locking_Range1;
5) Issue a Write command in Locking_Range1;
6) Invoke the `Set` method to configure Locking_Range1 ReadLockEnabled and WriteLockEnabled column values = TRUE, ReadLocked column value = TRUE and WriteLocked column values = FALSE;
7) Issue a Read command in Locking_Range1;
8) Issue a Write command in Locking_Range1;
9) Invoke the `Set` method to configure Locking_Range1 ReadLockEnabled and WriteLockEnabled column values = TRUE, ReadLocked column value = TRUE and WriteLocked column values = TRUE;
10) Issue a Read command in Locking_Range1; and
11) Issue a Write command in Locking_Range1.

Expected Response

1) Steps #1-3 SUCCEED;

 © TCG 2025

2) The commands issued in Step #4 SUCCEED. The commands issued in Step #5 FAIL;
3) Step #6 SUCCEED;
4) The commands issued in Step #7 FAIL. The commands issued in Step #8 SUCCEED;
5) Step #9 SUCCEED;
6) The commands issued in Step #10 and Step #11 all FAIL; and
7) For all supported Write commands in step #5, #8, and #11 and all supported Read commands in step #4, #7, and #10, the SD performs:
    a. For Read command, no data is transferred; and
    b. Return an interface status of Data Protection Error, (See [2]).

 © TCG 2025